Highlights

Information Budgets of Trajectory Health Indicators for Remaining Useful Life Prediction

Anonymous authors

- Certified information shows how much an HI keeps about remaining life
- Monotonicity scores can rise while the information in an HI vanishes
- Static scalar indicators cost up to 18% RMSE; supply level and velocity
- Ignoring operating regime or protocol raises RUL error by 8 to 26%
- Simple indicators with an age baseline match deep encoders on real data

Information Budgets of Trajectory Health Indicators for Remaining Useful Life Prediction

Anonymous authors

Abstract

Engineers choose health indicators (HIs) for remaining useful life (RUL) prediction by shape scores such as monotonicity, or by the accuracy of a predictor, without knowing how much information about the RUL an HI keeps. We show that the mean squared error of any RUL predictor is bounded below by the entropy power of the RUL given the HI. The bound yields a certificate: the held-out error of any predictor lower-bounds, in nats, the information that the HI retains beyond the age, and every 0.1 nat lowers the bound on the root mean squared error (RMSE) by about 10 percent. Four results give practical rules. A squeezed HI scores higher on monotonicity while losing information; a static scalar linear HI inflates the bound by a factor set by the angle to the optimal weights, which rotate with remaining life; a regime-blind read-out has an error floor; and an integrating encoder accumulates error that an observation correction removes. On 77 to 260 turbofan engines and lithium-ion cells the certificate is informative and passes a negative control. Squeezing raised the monotonicity from 0.13–0.24 to 0.41–0.58 on engines while the certified information fell to about zero, a static scalar HI cost up to 18 percent RMSE in one cell collection and at most 7 percent elsewhere, and ignoring the operating regime or protocol raised the error by 8 to 26 percent. No learned encoder beat simple HIs under the same age-fused predictor.

Keywords: information budget, health indicator, remaining useful life, information theory, information bottleneck, prognostics

1. Introduction

Prognostics and health management (PHM) predicts the remaining useful life (RUL) of components such as batteries, bearings and aero-engines, and drives condition-based maintenance: when to replace a cell, schedule a bearing change or ground an engine [1–4]. Methods are conventionally divided into physics-based, data-driven and hybrid [3, 5], and recent work is dominated by deep networks that map sensor histories to the RUL end to end [6, 7]. An established alternative compresses the monitoring data into a scalar *health indicator* (HI) and extrapolates it to a failure threshold D_{fail}. An HI can be inspected by an engineer, the threshold gives the end of life (EOL) a physical meaning, and, when the HI is modelled as a stochastic process, the RUL is a first hitting time (FHT) with a known distribution [1, 8, 9]. Recent studies follow the same line [10–12], within the wider scope of reliability science [13, 14].

Three bodies of work bear on how an HI should be judged. First, HIs are usually selected by *geometric* criteria: Coble and Hines [15] formalised monotonicity, prognosability and trendability and optimised them with a genetic algorithm, and variants are common [16–18]. HIs may also be learned by recurrent networks [19], autoencoders [20, 21] or

regressions on engineered features [22]; in both cases an HI is judged by its shape or by the accuracy of the predictor built on it [23]. Second, probabilistic RUL prediction from an HI is mostly built on Wiener degradation, whose first passage time is inverse Gaussian [24–26], with measurement error [8, 27], random unit-specific drift [28, 29], nonlinear drift [30] or threshold regression [9], applied to batteries [27], bearings [31] and highly reliable products [32] (reviews: [1, 33]). The RUL uncertainty then has an aleatoric part (future randomness) and an epistemic part (incomplete knowledge of state and parameters) [34–37], which deep ensembles are a common way to represent [38]. Third, information theory [39, 40] enters learning mainly through the information bottleneck [41–46]; estimating information from samples is difficult [47–50], and limits on estimator accuracy are usually stated through Cramér–Rao bounds [51–53], of which the entropy-power bound used here is the information-theoretic counterpart [40] (Remark 3).

Engineers who build an HI face five practical questions that these bodies of work do not answer. Does the HI keep the information that a predictor needs? Is a higher monotonicity score a better HI? Is one scalar enough, or are level and velocity both needed? Must operating regimes be modelled? Is a learned encoder worth its cost? In the sources we searched (bibliographic databases and the references above; this is not an exhaustive survey) we did not find a bound on the RUL error that depends on the HI only through the information it retains, nor an account of what a *scalar, static* indicator, a regime-blind read-out or an integrating encoder loses. A companion study [54] analysed what a static, per-snapshot projection of a multichannel signal retains about the degradation state. Here we analyse the HI *trajectory*, with its level, velocity, between-unit variability and operating regime. Two further studies by the same authors address how many run-to-failure trajectories [55] and which charging-phase features [56] are needed; the present paper addresses how much *information* an HI can at best retain. Figure 1 shows the setting and where each result applies.

The paper answers these questions as follows.

1. **A certificate in nats** (Section 2). The entropy-power bound on the RUL error turns the held-out error of any predictor into a lower bound on the information that the HI retains beyond the age, and every 0.1 nat lowers the bound on the root mean squared error (RMSE) by about 10% (Theorem 1). For Wiener degradation the error splits into an irreducible floor and a part that a better HI can remove (Section 3, Theorem 2).
2. **Four ways in which an HI loses information**, each with its engineering consequence (Section 4): a squeeze loses information while monotonicity saturates, so geometric scores do not measure it; a static scalar linear HI inflates the bound by $1+\rho\sin^2\psi$ because the optimal weights rotate from velocity to level; a regime-blind read-out has an error floor; and an integrating encoder accumulates error that an observation correction removes.
3. **Tests on real data and a benchmark** (Sections 5 and 6). On turbofan engines (100 to 260 units per subset) and lithium-ion cells (77 to 169 units) the certificate is informative and the four effects are measured; learned encoders are compared with simple indicators under a conservative protocol.
4. **Engineering guidance** (Section 7): a decision table and a four-step workflow.

The ingredients are classical: the data processing inequality, the chain rules, the maximum-entropy property of the Gaussian and the entropy-power bound [40]; the

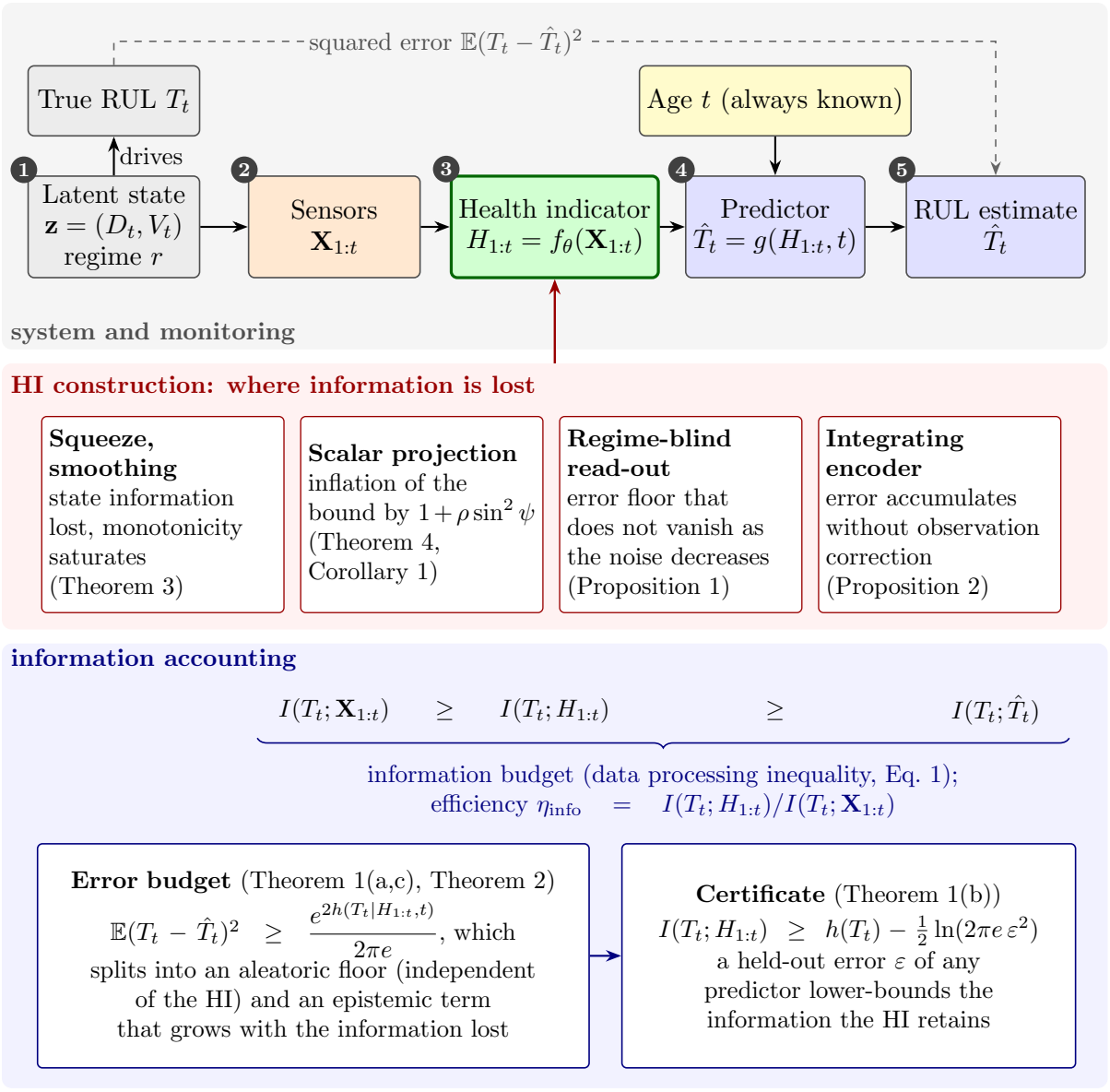


Figure 1: Framework of the paper. The HI is constructed from the sensors, which are driven by the same latent state as the true RUL, and the predictor uses the HI and the age. The data processing inequality gives an information budget; the red boxes list the operations of HI construction whose information loss is quantified.

inverse-Gaussian law [25]; Gaussian conditioning [57]; the Kalman filter [58, 59]; and the variational bottleneck bound [42]. What is new is their use as a budget for the HI-construction step, and the specific results listed above. Each proof states which classical facts it uses and where they can be found.

2. Information budget of a health indicator

Let $T_t > 0$ be the true RUL at inspection time (age) t, $\mathbf{X}_{1:t} \in \mathbb{R}^{d\times t}$ the accumulated history of d sensors, and $H_{1:t} = f_\theta(\mathbf{X}_{1:t})$ the HI trajectory produced by a map f_θ with parameters θ. Throughout, $h(\cdot)$ is differential entropy, $I(\cdot\,;\cdot)$ is mutual information (both in nats), $\mathbb{E}$ and Var are expectation and variance, and $\mathcal{N}(\mu,\sigma^2)$ is the normal distribution. Any RUL estimator $\hat{T}_t = g(H_{1:t})$ obeys the Markov chain $T_t \to \mathbf{X}_{1:t} \to H_{1:t} \to \hat{T}_t$, because

the latent degradation state drives both T_t and $\mathbf{X}_{1:t}$. The data processing inequality (DPI) [40, Sect. 2.8] gives

$$I(T_t; \mathbf{X}_{1:t}) \geq I(T_t; H_{1:t}) \geq I(T_t; \hat{T}_t), \tag{1}$$

with equality in the first inequality if and only if $H_{1:t}$ is a sufficient statistic of $\mathbf{X}_{1:t}$ for T_t [40, Sect. 2.9].

Theorem 1 (Information budget)**.** *Assume that $h(T_t)$ and $I(T_t; \mathbf{X}_{1:t})$ are finite, and let $\hat{T}_t$ be an estimator with finite second moment and mean squared error (MSE) $\varepsilon^2 = \mathbb{E}[(T_t - \hat{T}_t)^2]$.*

(a) Bound. *If $\hat{T}_t = g(H_{1:t})$, then*

$$\varepsilon^2 \;\geq\; \mathcal{B}_{\text{info}}(H_{1:t}) := \frac{1}{2\pi e} \exp\big(2h(T_t \mid H_{1:t})\big) = \frac{1}{2\pi e} \exp\big(2h(T_t) - 2I(T_t; H_{1:t})\big), \tag{2}$$

$\mathcal{B}_{\text{info}}(H_{1:t}) \geq \mathcal{B}_{\min} := \frac{1}{2\pi e} e^{2h(T_t|\mathbf{X}_{1:t})}$, *and*

$$\mathcal{B}_{\text{info}}(H_{1:t})/\mathcal{B}_{\min} = \exp\big(2\,(1-\eta_{\text{info}}(H_{1:t}))\, I(T_t; \mathbf{X}_{1:t})\big), \qquad \eta_{\text{info}}(H_{1:t}) := \frac{I(T_t; H_{1:t})}{I(T_t; \mathbf{X}_{1:t})} \in [0, 1]. \tag{3}$$

(b) Certificate. *Such an estimator certifies*

$$I(T_t; H_{1:t}) \;\geq\; h(T_t) - \tfrac{1}{2} \ln\big(2\pi e\, \varepsilon^2\big). \tag{4}$$

(c) Age. *The age t is always available to a predictor. If $\hat{T}_t = g(H_{1:t}, t)$, then $\varepsilon^2 \geq \frac{1}{2\pi e} \exp\big(2h(T_t \mid H_{1:t}, t)\big)$, and the information that the HI adds to the age satisfies*

$$I(T_t; H_{1:t}, t) = I(T_t; t) + I(T_t; H_{1:t} \mid t), \qquad I(T_t; H_{1:t} \mid t) \geq \mathbb{E}_t\, h(T_t \mid t) - \tfrac{1}{2} \ln\big(2\pi e\, \varepsilon^2\big). \tag{5}$$

Proof. (a) Let $E = T_t - g(H_{1:t})$. Differential entropy is translation invariant [40, Sect. 8.6], so $h(E \mid H_{1:t}) = h(T_t \mid H_{1:t})$ ($g(H_{1:t})$ is constant given $H_{1:t}$). Conditioning does not increase entropy [40, Sect. 2.6], so $h(E \mid H_{1:t}) \leq h(E)$, and the Gaussian maximises entropy at fixed second moment [40, Sect. 8.6], so $h(E) \leq \frac{1}{2} \ln(2\pi e\, \mathbb{E}[E^2])$. Combining the three relations gives $\mathbb{E}[E^2] \geq \frac{1}{2\pi e} e^{2h(T_t|H_{1:t})}$. The identity $h(T_t \mid H_{1:t}) = h(T_t) - I(T_t; H_{1:t})$ (definition of mutual information) gives the second form. By (1), $h(T_t \mid H_{1:t}) \geq h(T_t \mid \mathbf{X}_{1:t})$, and the ratio follows from $h(T_t \mid H_{1:t}) - h(T_t \mid \mathbf{X}_{1:t}) = I(T_t; \mathbf{X}_{1:t}) - I(T_t; H_{1:t})$. (b) Rearrange the first form of (a). (c) Apply (a) conditionally on t and average. The chain rule of mutual information [40, Sect. 2.5] gives the decomposition. For the inequality, (b) conditionally on t yields $I(T_t; H_{1:t} \mid t = \tau) \geq h(T_t \mid t = \tau) - \frac{1}{2} \ln(2\pi e\, \varepsilon_\tau^2)$ with ε_τ^2 the conditional MSE; averaging over τ and using Jensen's inequality for the concave logarithm [40, Sect. 2.6], $\mathbb{E}_\tau \ln \varepsilon_\tau^2 \leq \ln \mathbb{E}_\tau \varepsilon_\tau^2 = \ln \varepsilon^2$, gives the stated bound. □

Theorem 1 is the classical entropy-power bound on the error of an estimator [40, Sect. 8.6], used here to turn HI construction into an information budget; wherever the Bayesian Cramér–Rao bound applies, it is not weaker (Remark 3 in Appendix A).

Reading the theorem. Because $\mathcal{B}_{\text{info}} = e^{2h(T_t)-2I}/(2\pi e)$, an HI that retains I nats about the RUL cannot support an RMSE below e^{-I} times the bound for an HI that retains none: every 0.1 nat lowers the bound on the RMSE by about 10%. Part (b) turns the held-out MSE of any predictor into a number of nats, the *certificate*, which needs only the one-dimensional entropy $h(T_t)$, ranks HIs in the same order as the MSE and is vacuous

(negative) when the predictor is poor. Part (c) gives the relevant baseline: since the age is always known, the reference for an HI is the RUL given the age, not the marginal RUL, and a sensor-free predictor that uses only the age is a necessary baseline and a negative control, because it cannot certify information beyond the age (Appendix C). If the predictor uses the age, the certificate (b) applies to the HI trajectory together with its length t and already contains the age; the difference between the certificates of an HI-and-age predictor and of an age-only predictor is a difference of two lower bounds and certifies nothing about the information that the HI adds, which only (c) does. The efficiency η_{info} could in principle be estimated before any predictor is trained, but mutual-information estimates are unreliable on small data sets [49, 50] (Appendix E).

3. A physics-constrained error budget

Assume that after inspection the latent degradation follows $dD_s = \mu\, ds + \sigma_B\, dW_s$, $\mu > 0$ unit-specific (the random-effect formulation of [28, 29]), with absorbing threshold D_{fail}. The prognostically relevant latent state is $\mathbf{z} = (D_t, \mu)$, where the inspection level D_t is *not* observed directly but only through $\mathbf{X}_{1:t}$. With $a = D_{\text{fail}} - D_t > 0$, the RUL is conditionally inverse-Gaussian [24, 26], $T_t \mid \mathbf{z} \sim \mathcal{IG}(m(\mathbf{z}),\, a^2/\sigma_B^2)$ with mean $m(\mathbf{z}) = a/\mu$ and variance $a\sigma_B^2/\mu^3$ [25], and T_t is independent of $\mathbf{X}_{1:t}$ (hence of $H_{1:t}$) given $\mathbf{z}$. Here $\mathcal{IG}(\text{mean}, \text{shape})$ denotes the inverse Gaussian distribution. Define the information efficiency about the mean RUL $m = m(\mathbf{z})$ as $\eta_m(H) := I(m; H_{1:t})/I(m; \mathbf{X}_{1:t}) \in [0, 1]$.

Theorem 2 (Physics-constrained bound)**.** *Assume $h(m \mid \mathbf{X}_{1:t})$ and $I(m; \mathbf{X}_{1:t})$ are finite. For any estimator $\hat{T}_t = g(H_{1:t})$ with finite second moment,*

$$\mathbb{E}[(T_t - \hat{T}_t)^2] \;\geq\; \underbrace{\mathbb{E}\left[\frac{a\,\sigma_B^2}{\mu^3}\right]}_{\mathcal{E}_{\text{aleatoric}}} + \underbrace{\mathbb{E}_H\left[\frac{e^{2h(m|H_{1:t}=h)}}{2\pi e}\right]}_{\mathcal{E}_{\text{epistemic}}}, \tag{6}$$

and the epistemic term obeys

$$\mathcal{E}_{\text{epistemic}} \;\geq\; \frac{1}{2\pi e} \exp\Big(2h(m \mid \mathbf{X}_{1:t}) + 2\big(1 - \eta_m(H)\big)\, I(m; \mathbf{X}_{1:t})\Big). \tag{7}$$

The aleatoric term does not depend on the HI.

Proof. Since $\mathbb{E}[T_t - m(\mathbf{z}) \mid \mathbf{z}, H_{1:t}] = 0$ and T_t is conditionally independent of $H_{1:t}$ given $\mathbf{z}$, the cross term vanishes and $\mathbb{E}[(T_t - \hat{T}_t)^2] = \mathbb{E}[\mathrm{Var}(T_t \mid \mathbf{z})] + \mathbb{E}[(m(\mathbf{z}) - \hat{T}_t)^2]$, the usual split of the squared error into an irreducible and an estimation part [57, Sect. 1.5.5]. The first term is $\mathbb{E}[a\sigma_B^2/\mu^3]$. For the second, apply Theorem 1 conditionally on $H_{1:t} = h$ to the target $m(\mathbf{z})$, which gives $\mathbb{E}[(m - \hat{T}_t)^2 \mid H = h] \geq e^{2h(m|H=h)}/(2\pi e)$, and average over h. For the second inequality, the exponential is convex, so (Jensen [40, Sect. 2.6]) $\mathbb{E}_H[e^{2h(m|H=h)}] \geq e^{2h(m|H)}$ with $h(m \mid H) = \mathbb{E}_H h(m \mid H = h)$; and $m \to \mathbf{X}_{1:t} \to H_{1:t}$ is Markov, so $h(m \mid H) = h(m) - I(m; H) = h(m \mid \mathbf{X}) + I(m; \mathbf{X}) - I(m; H) = h(m \mid \mathbf{X}) + (1 - \eta_m) I(m; \mathbf{X})$. □

Remark 1. *Conditioning on D_t and changing variables would give $h(m \mid H) = h(\mu \mid H) + \mathbb{E}\ln(a/\mu^2)$. In the constant-drift model, D_t/t is then a sufficient statistic for μ and the resulting bound no longer depends on the HI. The HI matters through the joint uncertainty about the level and the drift, which is why Theorem 2 is stated for the latent pair $\mathbf{z}$. The second inequality of the theorem (the Jensen form) is weaker than the first but depends on the HI only through η_m.*

Table 1: Physics-constrained bound. $\mathcal{E}_{\text{ale}}$, $\mathcal{E}_{\text{epi}}$: aleatoric and epistemic terms (cycles^2); I in nats; errors in cycles.

ς	η_m	$I(m;H)$ [nats]	$\mathcal{E}_{\text{ale}}$	$\mathcal{E}_{\text{epi}}$ (exact)	$\mathcal{E}_{\text{epi}}$ (pointwise)	$\mathcal{E}_{\text{epi}}$ (Jensen)	$\sqrt{\text{bound}}$ [cycles]	Bayes RMSE [cycles]
0	1.00	1.40	63.9	86.6	75.7	34.3	11.81	12.26
2	0.96	1.35	64.0	91.2	80.1	37.9	12.00	12.46
4	0.91	1.28	63.4	99.0	86.6	44.1	12.25	12.74
8	0.80	1.13	63.8	121.5	106.4	59.7	13.05	13.61
16	0.63	0.88	62.9	174.7	150.9	97.3	14.62	15.42
32	0.38	0.53	63.8	318.7	265.0	198.6	18.13	19.56

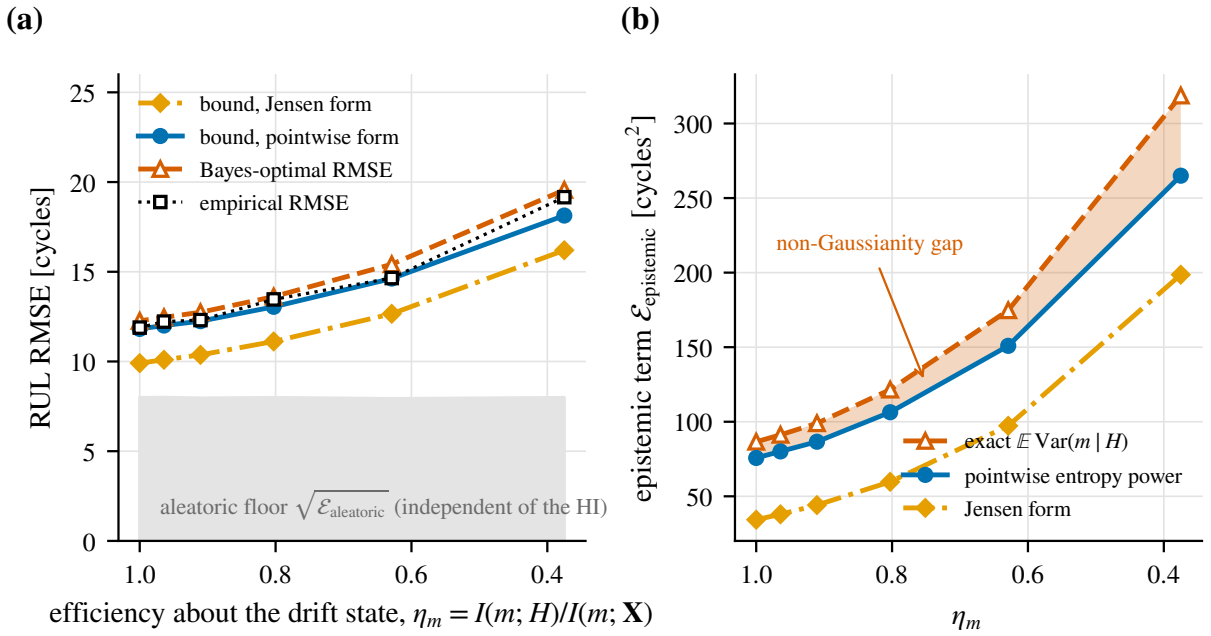


Figure 2: Physics-constrained bound as the HI loses information (decreasing η_m). (a) RMSE of the bounds and estimators. (b) Epistemic term; the shaded area is the non-Gaussianity gap.

The bound of Theorem 2 is conservative by an exact non-Gaussianity gap, the variance-weighted Kullback–Leibler (KL) divergence of the posterior of the mean RUL from a Gaussian (Proposition 5 in Appendix A).

Simulation check. The drift is $\mu \sim \mathcal{N}(1, 0.22^2)$, truncated to $\mu > 0.3$ and $a > 1$ (which keeps posterior sampling exact), with $\sigma_B = 0.85$, $D_{\text{fail}} = 100$ and $t = 50$. The raw sensor is $X_k = D_k + \xi_k$ with $\xi_k \sim \mathcal{N}(0, \sigma_\xi^2)$ and $\sigma_\xi = 1.5$. HIs with decreasing information are $H_k^\varsigma = X_k + \varsigma\zeta_k$ with independent standard normal ζ_k and $\varsigma \in \{0, 2, 4, 8, 16, 32\}$; each is a degraded version of X. The posterior of (D_t, μ) given H^ς is Gaussian (Kalman filter on (D, μ)), and $m = a/\mu$ is evaluated on 1500 posterior draws for each of 1500 test units; $h(m \mid H)$ and $h(m)$ are estimated with the Vasicek estimator. The Bayes-optimal error is $\mathbb{E}[\text{Var}(T_t \mid \mathbf{z})] + \mathbb{E}[\text{Var}(m \mid H)]$, and the empirical error against simulated first-passage times is also reported. Errors are given as RMSE.

The aleatoric term is constant (≈ 63.8 cycles^2), whereas the epistemic term grows as η_m falls from 1 to 0.38 (Table 1, Figure 2). The pointwise bound stays below the Bayes-optimal error at every noise level (RMSE 11.8 against 12.3 cycles for $\varsigma = 0$, and 18.1 against 19.6 for $\varsigma = 32$). The exact epistemic variance exceeds the pointwise value

because the posterior of m is not Gaussian; the mean KL divergence in Proposition 5 is about 0.07 nats at $\varsigma = 0$. The marginal density (B.2) of Appendix B, which mixes the inverse-Gaussian law over the Gaussian drift, agrees with numerical integration of the mixture to within 0.31%. The entropy estimates carry Monte Carlo error that we did not quantify, so gaps of a few per cent should not be over-interpreted.

Engineering consequence. The aleatoric term is a floor that no HI can lower. In the example it is 64 cycles^2 against 87 cycles^2 of epistemic error for a lossless HI, so that about 42% of the squared error is irreducible; comparing the achieved error of a predictor with this floor shows how much a better HI could still gain, and the epistemic term grows to 319 cycles^2 when the HI keeps only 38% of the information in the sensors.

4. Four ways in which an HI loses information

An HI can lose information about the RUL in four ways that are common in practice: by a squeeze that raises a geometric score (Section 4.1), by projection of the latent state on a static scalar (Section 4.2), by ignoring the operating regime (Section 4.3) and by integrating an estimated velocity without correction from the observations (Section 4.4). Each is stated as a result and checked on an exactly solvable model.

4.1. Squeezing: geometric criteria do not measure information

The geometric criteria of Coble and Hines [15] are computed from the shape of the HI. Monotonicity is $\mathrm{Mon}(H_{1:t}) = |\sum_{k=1}^{t-1} \mathrm{sgn}(H_{k+1} - H_k)|/(t-1)$, the balance of positive and negative increments. Prognosability is $\mathrm{Pro} = \exp(-\mathrm{std}(H_{\mathrm{end}})/\mathrm{mean}|H_{\mathrm{end}} - H_{\mathrm{start}}|)$, where the standard deviation and the mean are taken over units, H_{start} and H_{end} being the first and last values of each unit. In the benchmarks, Tre is the absolute correlation of the HI with time, as in [16] (and, under the name correlation, in [17]); it differs from the trendability of Coble and Hines, which measures the agreement of the functional form across units [15].

A strictly monotone bijection ϱ satisfies $I(T_t; \varrho(H_{1:t})) = I(T_t; H_{1:t})$, so information can be lost only through non-injective operations or noise. We therefore consider $\tilde{H}_{1:t} = \mathcal{S}(H_{1:t}, \xi)$, where ξ is independent of everything else. We call $\mathcal{S}$ a *squeeze* when it contracts the HI towards a template trajectory.

Theorem 3 (Squeezing)**.** *(a) For any such $\mathcal{S}$, $T_t \to H_{1:t} \to \tilde{H}_{1:t}$ is Markov, so $I(T_t; \tilde{H}_{1:t}) \le I(T_t; H_{1:t})$ and*

$$\mathcal{B}_{\mathrm{info}}(\tilde{H}_{1:t})/\mathcal{B}_{\mathrm{info}}(H_{1:t}) = \exp\big(2[I(T_t; H_{1:t}) - I(T_t; \tilde{H}_{1:t})]\big) \ge 1. \tag{8}$$

(b) Let the HI be the latent degradation level D_k observed with Gaussian noise, $H_k = D_k + \xi_k^{(1)}$, where D_k is linear-Gaussian, and let the squeeze with strength $\gamma \in [0, 1)$ and template (mean trajectory) $m_k = \mathbb{E}[D_k]$ be

$$\begin{gathered} \tilde{H}_k^{\gamma} = m_k + (1-\gamma)(H_k - m_k) + \xi_k^{(2)}, \\ \xi_k^{(2)} \sim \mathcal{N}(0, \sigma_2^2) \text{ independent and identically distributed (i.i.d.)} \end{gathered} \tag{9}$$

For every linear functional ϑ of the latent state (for example the current velocity), $\mathcal{I}_{\mathrm{tot}}^{\vartheta}(\gamma) := I(\vartheta; \tilde{H}_{1:t}^{\gamma})$ is non-increasing in γ and

$$\mathcal{I}_{\mathrm{tot}}^{\vartheta}(\gamma) \xrightarrow[\gamma \to 1]{} 0. \tag{10}$$

Table 2: Squeezing experiment (mean over three seeds; ±: standard deviation). Information about V_t in nats, errors in cycles.

γ	Mon	$\mathcal{I}_{\text{state}}$ [nats]	$\mathcal{I}_{\text{kin}}$ [nats]	η_{info}	$\sqrt{\mathcal{B}_{\text{info}}}$ [cycles]	Bayes RMSE [cycles]	Empirical RMSE [cycles]
0.000	0.496	0.789	0.092	1.00	6.67	8.04	15.0±9.8
0.500	0.766	0.779	0.092	0.99	6.81	8.91	14.9±9.4
0.900	0.975	0.567	0.188	0.85	7.38	9.27	15.3±9.2
0.970	0.981	0.159	0.396	0.63	9.01	10.99	17.1±9.2
0.990	0.981	0.022	0.200	0.25	12.87	15.20	21.2±8.9
0.995	0.982	0.006	0.070	0.09	14.88	17.83	23.2±8.9

If the template has constant increment Δm, then as $t \to \infty$ and $\gamma \to 1$,

$$\text{Mon}(\tilde{H}^{\gamma}_{1:t}) \to \text{Mon}_{\infty} = 2\Phi_{\mathcal{N}}\big(\Delta m/(\sqrt{2}\,\sigma_2)\big) - 1, \tag{11}$$

where $\Phi_{\mathcal{N}}$ is the standard normal distribution function; the limit does not depend on the unit's own degradation.

Proof sketch. Part (a) follows from the data processing inequality. In (b) the squeezed HI is an invertible affine image of H observed through Gaussian noise whose variance grows without bound as $\gamma \to 1$, so the information vanishes, and the limit of Mon follows from the strong law of large numbers for the 1-dependent increments. The full proof is in Appendix A. □

Part (b) shows that the geometric score and the information content are decoupled. Once the squeeze is strong, Mon saturates at a value set by the template and the noise, while the information available for RUL prediction keeps falling and the bound $\mathcal{B}_{\text{info}}$ keeps rising. The loss splits into a part about the current state and a part about the kinematics (Proposition 4 in Appendix A).

Simulation check. The latent state is a local linear trend, $D_k = D_{k-1} + V_{k-1} + \sigma_B w_k$ and $V_k = V_{k-1} + \sigma_V \zeta_k$, with level D_k, velocity V_k, independent standard normal noises w_k, ζ_k and $V_0 \sim \mathcal{N}(\bar{v}, \sigma_{v0}^2)$. The raw HI is $H_k = D_k + \xi_k^{(1)}$ with measurement noise $\xi_k^{(1)} \sim \mathcal{N}(0, \sigma_1^2)$, and the squeeze is that of Theorem 3(b) with noise standard deviation σ_2. The parameters are $\bar{v} = 2$, $\sigma_{v0} = 0.25$, $\sigma_V = 0.03$, $\sigma_B = 0.5$, $\sigma_1 = 2$, $\sigma_2 = 0.6$, failure threshold $D_{\text{fail}} = 200$ and inspection at $t = 50$. The information about V_t is exact from the Kalman covariances [58, 59]. For the RUL, the Bayes-optimal estimator is the posterior mean from 300 exact posterior draws per unit, $h(T_t \mid \tilde{H})$ is estimated with the Vasicek estimator [47], and $\sqrt{\mathcal{B}_{\text{info}}}$ is the square root of the average of the pointwise entropy powers. Errors are reported as RMSE; each setting uses 400 test units and three seeds.

Monotonicity rises from 0.50 to 0.98 and saturates at the predicted value $\text{Mon}_{\infty} = 0.981$ for $\gamma \geq 0.97$ (Table 2, Figure 3), while from $\gamma = 0.9$ to 0.995 the efficiency falls from 0.85 to 0.09 and the Bayes-optimal RMSE nearly doubles (9.3 to 17.8 cycles). The bound stays below the Bayes-optimal error in every run. The state information falls monotonically, whereas the kinematic information first rises and then falls, so the inflation at $\gamma = 0.9$ and 0.97 is driven by the loss of state information. The empirical RMSE varies strongly between seeds because of the heavy right tail of the RUL, and the posterior-based Bayes RMSE is the reliable summary.

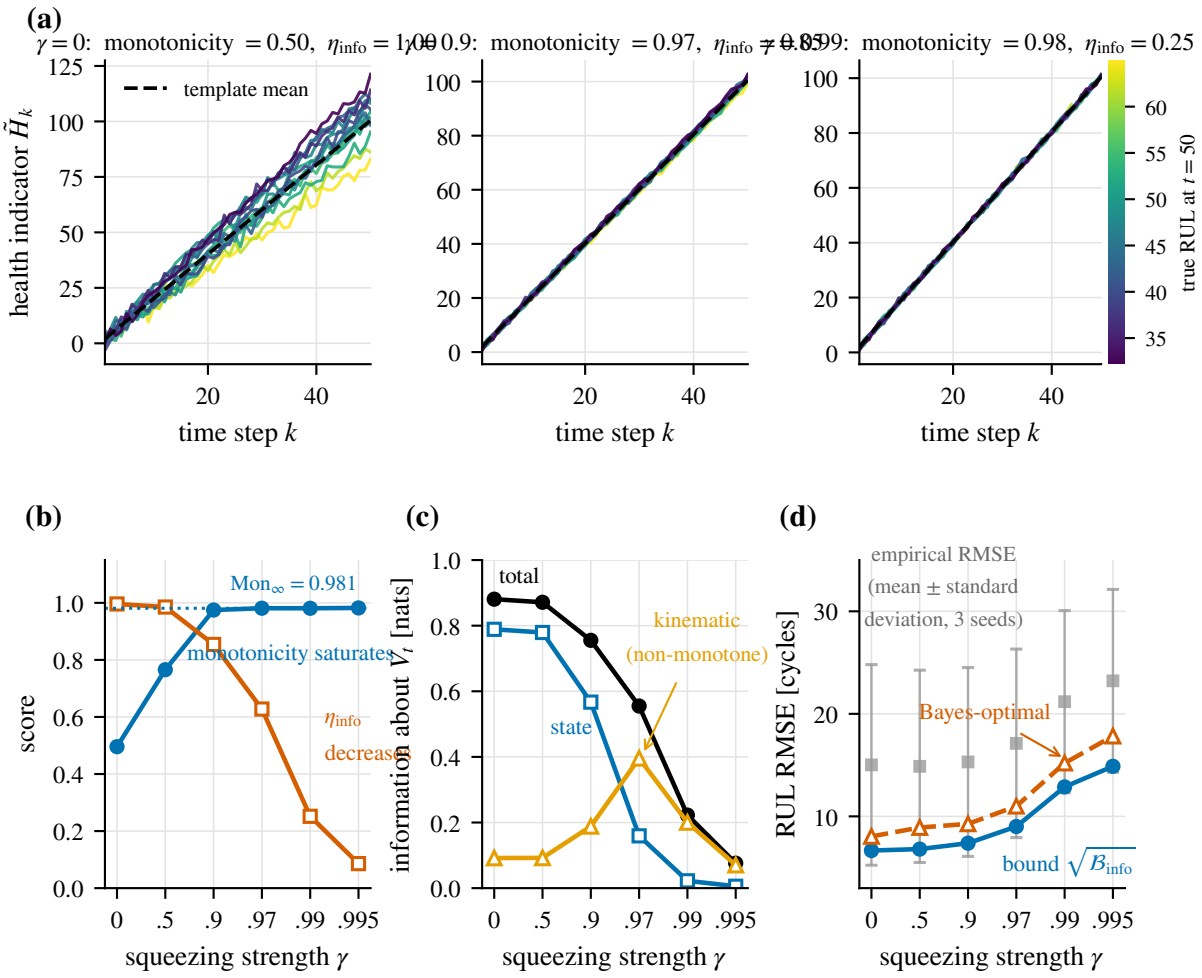


Figure 3: Squeezing experiment. (a) HI paths of 14 units, coloured by the true RUL. (b) Monotonicity and efficiency. (c) Information about V_t. (d) Bound and RMSE (error bars: standard deviation over seeds). The values of γ are equally spaced.

Engineering consequence. Monotonicity, trendability and prognosability should not be used to select or tune an HI: an HI can be made to score higher by squeezing it toward a template while it loses information. Rank candidate HIs by certified information instead (Section 5).

4.2. Projection: scalar indicators at the trajectory level

Theorem 2 concerns the information budget of an arbitrary HI. We now ask what a *scalar*, *static* HI loses when the degradation velocity varies in time. The loss has the same $\sin^2$ structure as the projection-misalignment loss at the snapshot level in [54], but it acts along the *temporal* direction (level versus velocity).

Setting. Let $\mathbf{z} = (D_t, V_t)^\top$ be the level and the velocity at inspection time t, with Gaussian prior $\mathcal{N}(\mathbf{z}_0, \boldsymbol{\Sigma})$, observed through $\mathbf{X}_{1:t}$ in a linear-Gaussian model (for example the local-linear-trend state space [59, 60] of Section 4.1 with Gaussian sensor noise). The posterior is $\mathcal{N}(\hat{\mathbf{z}}, \mathbf{P})$ where $\hat{\mathbf{z}} = \mathbb{E}[\mathbf{z} \mid \mathbf{X}_{1:t}]$ and $\mathbf{P}$ does not depend on the data (as for the covariance recursion of the Kalman filter [58, 59]); $\hat{\mathbf{z}} \sim \mathcal{N}(\mathbf{z}_0, \boldsymbol{\Lambda})$ with $\boldsymbol{\Lambda} = \boldsymbol{\Sigma} - \mathbf{P} \succ 0$ (law of total variance [57, Sect. 1.5.5]), and $\hat{\mathbf{z}}$ is a sufficient statistic of $\mathbf{X}_{1:t}$ for $\mathbf{z}$. The target is the frozen-velocity RUL $m(\mathbf{z}) = (D_{\mathrm{fail}} - D_t)/V_t$, which is nonlinear in $\mathbf{z}$. We linearise it at

a *fixed*, non-random operating point $\bar{\mathbf{z}} = (\bar{D}, \bar{V})^\top$, for example the prior mean at the inspection stage. Then

$$\mathbf{g} = \nabla m(\bar{\mathbf{z}}) = -\frac{1}{\bar{V}}\Big(1,\ \frac{\bar{a}}{\bar{V}}\Big)^\top, \qquad \bar{a} = D_{\text{fail}} - \bar{D}, \tag{12}$$

is a fixed vector, and $m_L(\mathbf{z}) = m(\bar{\mathbf{z}}) + \mathbf{g}^\top(\mathbf{z} - \bar{\mathbf{z}})$ is a fixed linear functional of $\mathbf{z}$. All statements of this section are exact for m_L; the error of replacing m by m_L is not covered by the theorems and is examined numerically in Section 4.2. A scalar *linear* HI built from the sufficient statistic, with fixed weights $\mathbf{u}$, is $h_{\mathbf{u}} = \mathbf{u}^\top \hat{\mathbf{z}}$. Define the signal-to-noise ratio $\rho = \mathbf{g}^\top \mathbf{\Lambda} \mathbf{g} / \mathbf{g}^\top \mathbf{P} \mathbf{g}$ and the $\mathbf{\Lambda}$-angle $\psi \in [0, \pi/2]$ between $\mathbf{g}$ and $\mathbf{u}$,

$$\cos\psi = \frac{|\mathbf{g}^\top \mathbf{\Lambda} \mathbf{u}|}{\sqrt{(\mathbf{g}^\top \mathbf{\Lambda} \mathbf{g})(\mathbf{u}^\top \mathbf{\Lambda} \mathbf{u})}}. \tag{13}$$

Theorem 4 (Angular loss law)**.** *In the setting above, for every fixed* $\mathbf{u} \neq 0$*:*
(a) $\operatorname{Var}(m_L \mid h_{\mathbf{u}}) = \mathbf{g}^\top \mathbf{P} \mathbf{g} + (\mathbf{g}^\top \mathbf{\Lambda} \mathbf{g}) \sin^2 \psi$.
(b) For the linearised target, with $I(m_L; \mathbf{X}_{1:t}) = \frac{1}{2}\ln(1+\rho)$*,*

$$\frac{\mathcal{B}_{\text{info}}(h_{\mathbf{u}})}{\mathcal{B}_{\min}} = 1 + \rho \sin^2 \psi, \qquad \eta_m(h_{\mathbf{u}}) = 1 - \frac{\ln(1 + \rho \sin^2 \psi)}{\ln(1+\rho)}. \tag{14}$$

(c) $\eta_m(h_{\mathbf{u}}) = 1$ *if and only if* $\mathbf{u} \parallel \mathbf{g}$*; the HI* $h_{\mathbf{g}} = \mathbf{g}^\top \hat{\mathbf{z}}$ *is information-lossless for* m_L *and is, by Lemma 1 (Appendix A), the best scalar linear HI of* $\mathbf{X}_{1:t}$*.*

Proof. (a) Because $\mathbf{z} = \hat{\mathbf{z}} + \boldsymbol{\varepsilon}$ with $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \mathbf{P})$ independent of $\hat{\mathbf{z}}$, we have $\operatorname{Cov}(\mathbf{z}, \hat{\mathbf{z}}) = \mathbf{\Lambda}$ and $\operatorname{Var}(\mathbf{g}^\top \mathbf{z}) = \mathbf{g}^\top \mathbf{\Sigma} \mathbf{g} = \mathbf{g}^\top \mathbf{\Lambda} \mathbf{g} + \mathbf{g}^\top \mathbf{P} \mathbf{g}$. The pair $(\mathbf{g}^\top \mathbf{z}, h_{\mathbf{u}})$ is jointly Gaussian with $\operatorname{Cov} = \mathbf{g}^\top \mathbf{\Lambda} \mathbf{u}$ and $\operatorname{Var}(h_{\mathbf{u}}) = \mathbf{u}^\top \mathbf{\Lambda} \mathbf{u}$, so the Gaussian conditioning formula [57, Sect. 2.3] gives $\operatorname{Var}(m_L \mid h_{\mathbf{u}}) = \mathbf{g}^\top \mathbf{\Lambda} \mathbf{g} + \mathbf{g}^\top \mathbf{P} \mathbf{g} - (\mathbf{g}^\top \mathbf{\Lambda} \mathbf{u})^2 / (\mathbf{u}^\top \mathbf{\Lambda} \mathbf{u}) = \mathbf{g}^\top \mathbf{P} \mathbf{g} + \mathbf{g}^\top \mathbf{\Lambda} \mathbf{g}\,(1 - \cos^2 \psi)$.
(b) In the linearised model the conditional law of m_L is Gaussian, so the entropy power equals the variance. Taking the ratio of (a) to the case $h = \hat{\mathbf{z}}$, for which $\operatorname{Var}(m_L \mid \hat{\mathbf{z}}) = \mathbf{g}^\top \mathbf{P} \mathbf{g}$, gives the first formula. Moreover, by the formula for the mutual information of jointly Gaussian variables [40, Chap. 8], $I(m_L; \mathbf{X}) = \frac{1}{2}\ln(\operatorname{Var}(m_L)/\mathbf{g}^\top \mathbf{P} \mathbf{g}) = \frac{1}{2}\ln(1+\rho)$ and $I(m_L; h_{\mathbf{u}}) = \frac{1}{2}\ln\big((1+\rho)/(1+\rho\sin^2\psi)\big)$; their ratio is η_m.
(c) $\sin^2 \psi = 0$ iff $\mathbf{\Lambda}^{1/2}\mathbf{g} \parallel \mathbf{\Lambda}^{1/2}\mathbf{u}$ iff $\mathbf{u} \parallel \mathbf{g}$ since $\mathbf{\Lambda} \succ 0$. For $\mathbf{u} = \mathbf{g}$ the loss vanishes, and Lemma 1 shows that no scalar linear functional of $\mathbf{X}$ can do better. □

Remark 2 (Scope of the angular law)**.** *The theorem concerns scalar* linear *HIs with fixed weights and the linearised target. If the operating point is replaced by the posterior mean* $\hat{\mathbf{z}}$ *of each unit, the gradient becomes a function of the data, and the theorem then describes the loss conditionally on freezing the gradient, that is, a local analysis around the unit's own estimate. An HI with unit-specific weights is a nonlinear scalar function of* $\hat{\mathbf{z}}$ *and is not covered by the optimality statements, which also do not exclude nonlinear scalar HIs with fixed structure.*

Corollary 1 (No static scalar linear HI is optimal over the whole life)**.** *(a) In the coordinates* (D_t, V_t) *the optimal direction is* $\mathbf{g}_t \propto (1, \bar{\tau}_t)^\top$*, with* $\bar{\tau}_t = \bar{a}_t / \bar{V}_t$ *the frozen-velocity RUL (in steps) at the operating point of stage* t*. The optimal scalar linear HI therefore weights the velocity estimate* $\bar{\tau}_t$ *times more than the level estimate; the weight moves from the velocity (early life,* $\bar{\tau}_t$ *large) to the level (end of life,* $\bar{\tau}_t \to 0$*), and the Euclidean angle*

Table 3: Inflation of the bound at $t = 50$, mean over 500 units: prediction at the fixed operating point, prediction at the unit's own estimate, and Monte Carlo for the exact target. ψ: angle to the unit-specific gradient.

HI	mean ψ [deg]	fixed point	unit gradient	Monte Carlo
level only $\mathbf{u} = (1, 0)$	6.0	1.179	1.177	1.181
velocity only $\mathbf{u} = (0, 1)$	5.8	1.164	1.194	1.197
stage-fixed gradient (prior mean)	0.7	1.000	1.004	1.003
unit-specific gradient	0.0	1.004	1.000	0.999

between the optimal directions at RULs τ_1, τ_2 is $|\arctan\tau_1 - \arctan\tau_2|$.
(b) Let $\mathbf{g}_1, \mathbf{g}_2$ be the gradients at the operating points of two inspection stages with information matrices $\mathbf{\Lambda}_1, \mathbf{\Lambda}_2$ and ratios $\rho_1, \rho_2 > 0$. If $\mathbf{g}_1 \not\parallel \mathbf{g}_2$, then for every fixed $\mathbf{u}$ the bound is inflated at one of the two stages, $\max_j \rho_j \sin^2\psi_j(\mathbf{u}) \geq \delta > 0$; if $\mathbf{\Lambda}_1 = \mathbf{\Lambda}_2$, then $\delta \geq \min_j \rho_j \sin^2(\psi_{12}/2)$ with ψ_{12} the $\mathbf{\Lambda}$-angle between $\mathbf{g}_1$ and $\mathbf{g}_2$. A stage-wise HI that uses the weights $\mathbf{g}_j$ at stage j has no inflation, relative to the linearised target, at any stage.

Proof sketch. Part (a) is the gradient formula above, and (b) follows from Theorem 4(c), which makes the inflation at stage j vanish only for $\mathbf{u} \parallel \mathbf{g}_j$, and from the triangle inequality for the $\mathbf{\Lambda}$-angle between lines. The full proof is in Appendix A. □

Corollary 1(a) explains why an indicator built from a level-like feature, for example the first principal component of cumulative-damage features, loses information early in life, and why a velocity-like feature loses it late in life. It is the temporal counterpart of the snapshot-level rotation of the signal direction. It also motivates the predictor of Appendix B.4, which uses both the smoothed HI level and its windowed slope: this pair estimates the two-dimensional sufficient statistic $\hat{\mathbf{z}}$.

Simulation check. The model is the local linear trend of Section 4.1 with the raw sensor and $D_{\text{fail}} = 200$. The posterior of $\mathbf{z} = (D_t, V_t)$ is Gaussian, and $\mathbf{\Lambda}$ is strongly anisotropic (eigenvalues 0.004 and 204). For each of 500 units, the exact entropy power of the *nonlinear* target $m = (D_{\text{fail}} - D_t)/V_t$ given $h_{\mathbf{u}}$ is estimated from the exact conditional Gaussian of $\mathbf{z}$ (2000 draws, Vasicek estimator) and compared with $1 + \rho\sin^2\psi$ computed in two ways: at a fixed operating point, the prior mean of the stage, as in Theorem 4; and at the unit's own posterior mean, a local analysis in the sense of Remark 2. As a sanity check, in the constant-drift model of Section 3 $\mathbf{\Lambda}$ is effectively of rank one and the predicted inflation is 1.000 for every $\mathbf{u}$ (Monte Carlo: 1.000 to 1.003), as the remark after Theorem 2 implies.

At $t = 50$ both predictions agree with the Monte Carlo value within about 3% for all HIs (Table 3). Figure 4 illustrates Corollary 1. The static HI aligned at $t^* = 50$ is lossless at $t = 25$ and 50 but inflates the bound by about 40% at $t = 75$ (Monte Carlo 1.39), the level-only HI by about 37% and the velocity-only HI by a factor of about 5. The prediction at the fixed prior-mean operating point agrees with Monte Carlo at $t = 25$ and 50 but underestimates the late-life inflation of the static and the velocity-only HI (1.12 and 3.5 at $t = 75$), because the gradient of the nonlinear target at the prior mean differs from that at the units' own estimates; the prediction with the unit's own gradient (1.40 and 5.1) agrees within 3%. The linear theory is therefore an accurate local approximation but not an exact statement for the nonlinear target.

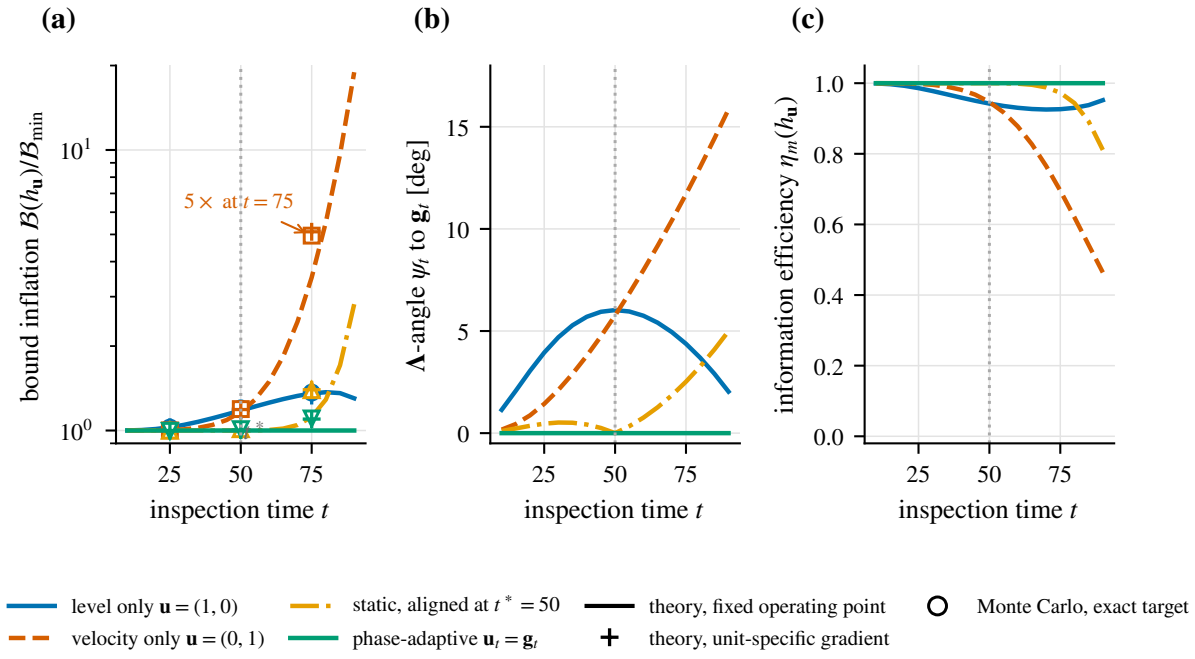


Figure 4: Angular loss law for scalar linear HIs. (a) Inflation of the bound. (b) Angle to the optimal direction $\mathbf{g}_t$ at the fixed operating point. (c) Information efficiency η_m. The static HI is aligned at $t^* = 50$; the phase-adaptive HI uses the stage-wise gradient at the prior mean.

Engineering consequence. A static scalar HI is adequate only when the stage-optimal directions are close. Otherwise pass the level and the velocity of the HI to the predictor, as the windowed slope does in our predictor, or let the weights follow the stage of life.

4.3. Regime blindness

When the sensors are recorded under several operating regimes, the sensor responds to the degradation with a regime-dependent offset and gain, whereas the degradation process itself is the same. The following statement quantifies what a regime-blind read-out loses.

Proposition 1 (Regime leakage floor)**.** *Let a latent level D with mean $\bar{d}$ and variance v be observed through a sensor as $x = a_r D + b_r + \varepsilon$, where the regime r takes R values with probabilities π_r independently of D and of the noise ε (variance σ^2); a_r and b_r are the gain and offset of regime r. Put $\bar{a} = \mathbb{E}a_r$, $V_x = \mathbb{E}[a_r^2]\mathbb{E}[D^2] - (\bar{a}\bar{d})^2 + \mathrm{Var}(b_r) + 2\bar{d}\,\mathrm{Cov}(a_r, b_r) + \sigma^2$. The best regime-blind linear read-out of D from x has mean squared error*

$$\mathrm{MSE}_{\mathrm{blind}} = v - \frac{\bar{a}^2 v^2}{V_x}, \qquad \textit{and} \qquad \mathrm{MSE}_{\mathrm{blind}} \xrightarrow[\sigma \to 0]{} \frac{v\,[\mathrm{Var}(a_r)\,v + \mathrm{Var}(\bar{d}a_r + b_r)]}{\bar{a}^2 v + \mathrm{Var}(a_r)\,v + \mathrm{Var}(\bar{d}a_r + b_r)} > 0 \tag{15}$$

unless the regimes agree in a_r and in $\bar{d}a_r + b_r$, whereas a read-out that knows the regime has $\mathrm{MSE}_{\mathrm{aware}} = \sigma^2 \sum_r \pi_r / a_r^2 \to 0$.

Proof. Because D and r are independent, $\mathrm{Cov}(D, x) = \bar{a}\,v$, and $\mathrm{Var}(x) = \mathrm{Var}(a_r D) + \mathrm{Var}(b_r) + 2\bar{d}\,\mathrm{Cov}(a_r, b_r) + \sigma^2 = V_x$, since $\mathrm{Cov}(a_r D, b_r) = \bar{d}\,\mathrm{Cov}(a_r, b_r)$ and $\mathrm{Var}(a_r D) = \mathbb{E}[a_r^2]\mathbb{E}[D^2] - (\bar{a}\bar{d})^2$; the best linear predictor of D from x has error $v - \mathrm{Cov}(D, x)^2 / \mathrm{Var}(x)$ (the conditional variance of a Gaussian pair with the same second moments [57, Sect. 2.3]). For the limit use $\mathbb{E}[a_r^2]\mathbb{E}[D^2] - (\bar{a}\bar{d})^2 = \bar{a}^2 v + \mathrm{Var}(a_r)(v + \bar{d}^2)$ and $\mathrm{Var}(a_r)\bar{d}^2 + \mathrm{Var}(b_r) +$

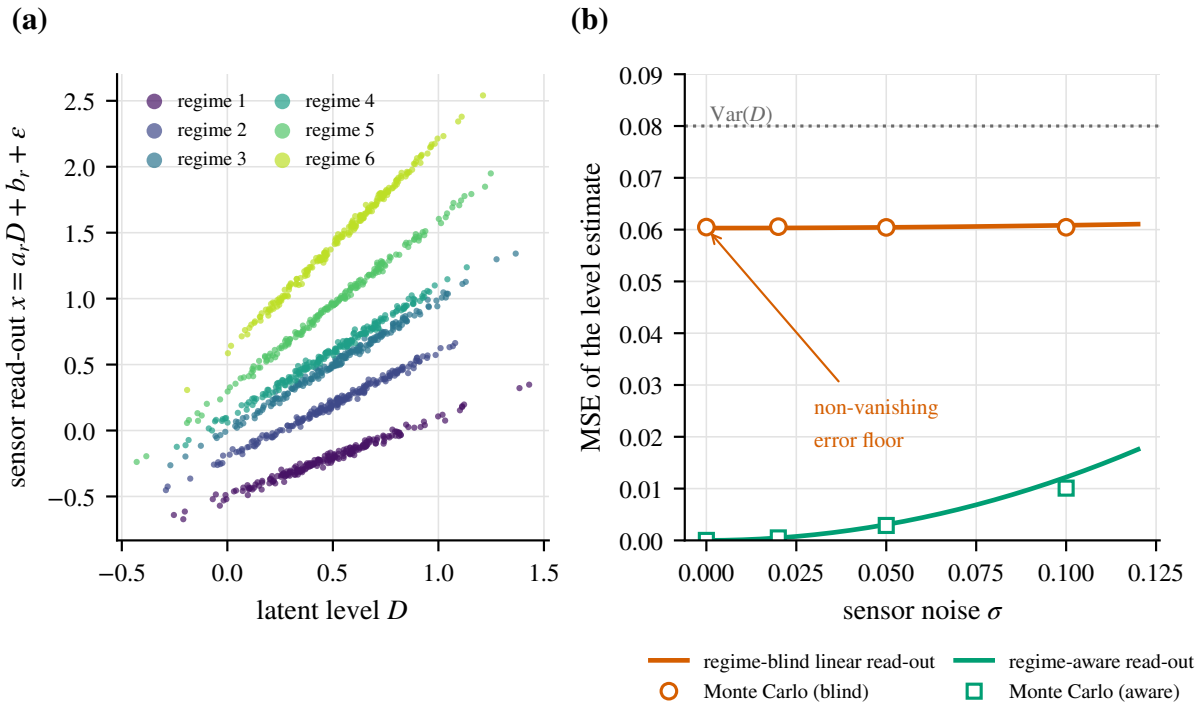


Figure 5: Regime leakage in the six-regime example. (a) Sensor read-out against latent level for each regime. (b) Error of the regime-blind and regime-aware read-outs: formula (lines) and simulation (markers).

$2\bar{d}\mathrm{Cov}(a_r, b_r) = \mathrm{Var}(\bar{d}a_r + b_r)$, so that at $\sigma = 0$, $V_x = \bar{a}^2 v + \mathrm{Var}(a_r)v + \mathrm{Var}(\bar{d}a_r + b_r)$. Given the regime, the read-out $(x - b_r)/a_r$ is unbiased with variance σ^2/a_r^2. □

Normalising the sensors within each regime removes the regime offsets but not the regime-dependent gains, so the floor remains. In a numerical example with six regimes ($a_r \in [0.6, 1.6]$, $b_r \in [-0.5, 0.6]$, $v = 0.08$), the formula and a simulation agree on $\mathrm{MSE}_{\mathrm{blind}} = 0.060$ at $\sigma = 0$, so that only 25% of $\mathrm{Var}(D)$ is explained, whereas the regime-aware read-out has an error that tends to zero (Figure 5).

Engineering consequence. Normalise the sensors by operating regime and, when protocols differ between units, by each unit's own early cycles. Per-regime normalisation removes offsets but not gains, so the regime should also be given to the observation model when gains differ.

4.4. Integration without observation correction

An HI that integrates an estimated velocity, $H_k = H_{k-1} + u_k$, has nothing that anchors its level to the observations. Section 4.2 treats the level and the velocity as latent quantities, and the level must be re-estimated from the sensors at every step; this is the measurement update of a Kalman filter [58]. An observation-corrected HI adds a level read-out l_k and a gain $\omega_k \in (0, 1)$, $H_k = (1 - \omega_k)(H_{k-1} + u_k) + \omega_k\, l_k$, and $\omega = 0$ is the pure integrator. The next proposition quantifies what the integrator accumulates.

Proposition 2 (Error accumulation without observation correction)**.** *Let the velocity estimate carry the error $e_j = b + w_j$ with a unit-specific persistent bias b and white noise w_j of variance σ_w^2, and let the level read-out carry independent zero-mean noise of variance*

σ_n^2. *For the pure integrator ($\omega = 0$) the level error after t steps has mean tb and variance $t\sigma_w^2$, both unbounded in t. For a constant gain $\omega \in (0, 1]$ the level error converges to a stationary law with*

$$\mathbb{E}\,\varepsilon_\infty = \frac{(1-\omega)\,b}{\omega}, \qquad \operatorname{Var}\,\varepsilon_\infty = \frac{(1-\omega)^2\sigma_w^2 + \omega^2\sigma_n^2}{\omega(2-\omega)}. \tag{16}$$

Proof. The level error obeys $\varepsilon_k = (1-\omega)(\varepsilon_{k-1} + e_k) + \omega n_k$ with $\varepsilon_0 = 0$; write $\alpha = 1 - \omega$. For $\omega = 0$, $\varepsilon_t = \sum_{j \le t} e_j$ has mean tb and variance $t\sigma_w^2$. For $\omega > 0$ the mean satisfies $m_k = \alpha(m_{k-1} + b)$, with fixed point $\alpha b/(1-\alpha) = (1-\omega)b/\omega$ approached geometrically, and the variance $v_k = \alpha^2 v_{k-1} + \alpha^2\sigma_w^2 + \omega^2\sigma_n^2$ (independent noise terms) has fixed point $(\alpha^2\sigma_w^2 + \omega^2\sigma_n^2)/(1-\alpha^2)$ with $1 - \alpha^2 = \omega(2-\omega)$. □

A simulation ($b = 0.01$, $\sigma_w = 0.02$, $\sigma_n = 0.15$, 400 steps, 2×10^5 paths) agrees with both expressions (for $\omega = 0$ the bias is $4.00 = tb$; for $\omega = 0.2$ the bias and variance are 0.0399 and 0.00321 against 0.04 and 0.00321). A constant gain amounts to exponential smoothing, the steady state of the local-level filter [59, 60]. A persistent velocity bias is the generic situation when units differ in initial offset or wear, and it is the mechanism that we diagnose for the integrating encoder on the turbofan data (Section 6). Two learned encoders implement this correction with a learned gain: PGIB-KF, an observation-corrected (Kalman-filter-based) variant of the physics-guided information bottleneck (PGIB), and its regime-aware variant PGIB-KFR. They are described in Appendix B and compared with simple HIs in Section 6.

Engineering consequence. An encoder that integrates a velocity needs an observation correction: a gain on a level read-out turns an unbounded error into a bounded one.

5. Tests on real data

This section tests the predictions of Sections 2 to 4 directly, on two domains with many units: turbofan engines (the four subsets FD001 to FD004 of the Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) data, 100 to 260 training engines) and lithium-ion cells (the HUST, MATR and Tongji collections, with 77, 169 and 88 cells). Estimators are compared in Section 6.

5.1. Data and design

All analyses use unit-grouped five-fold cross-validation: preprocessing, HIs and predictors are fitted on the training units of a fold and evaluated on its held-out units. The population consists of pairs of a unit and an age t between a minimum age (30 cycles for engines, 100, 50 and 30 cycles for the HUST, MATR and Tongji cells) and the age at which 10% of the training units are still alive. Pooling all ages of all units gives $T_t \mid t$ the law of $L - t$ given $L > t$ for the lifetime L, so that $h(T_t \mid t)$ is estimated from the lifetime distribution of the training units (Gaussian kernel density of $\ln L$) without further assumption. Predictors are gradient-boosted trees on summaries of the HI trajectory and the age, and confidence intervals are bootstrap intervals over units (Appendix F).

Turbofan engines. The training sets of the four C-MAPSS subsets are used (FD001 and FD003: one operating regime; FD002 and FD004: six regimes), with 14 informative sensors standardised within regime where there are several. The HIs are a ridge regression of the life fraction on the standardised sensors (ridge HI) and the first principal component of the sensors (PCA-HI); a predictor on the raw sensors is the reference.

Lithium-ion cells. The HUST collection [61] contains 77 lithium iron phosphate/graphite cells, each cycled under its own multi-stage charge and discharge protocol, with end of life at 80% of the initial capacity (977 to 2334 cycles; coefficient of variation of the lifetime 0.19). The MATR collection [22, 62] contains 169 cells of the same chemistry cycled under fast-charging protocols, with end of life at 80% of the nominal capacity (147 to 2235 cycles; coefficient of variation 0.42). The Tongji collection [63] contains 88 cells (of 130 recorded) of three nickel-based chemistries (lithium nickel cobalt aluminium oxide (NCA), lithium nickel manganese cobalt oxide (NMC) and a blend, with graphite or graphite/silicon anodes), cycled at 25, 35 and 45 °C under several protocols, with end of life at 80% of the nominal capacity (121 to 570 cycles; coefficient of variation 0.37); the other cells did not reach this threshold within the recordings or lasted fewer than 100 cycles. The three collections were taken from the processed BatteryLife repository [64]. The HIs use only features of the charging phase (charge duration and charge voltage at 27% and 73% of the nominal capacity, that is 0.3 and 0.8 Ah for the 1.1 Ah cells), each expressed relative to cycles 2 to 11 of the same cell. The discharge capacity, which defines the end of life, enters only a reference predictor. Time scales are matched to the lifetimes (Appendix F).

5.2. Age-conditional certificate

Turbofan engines (Theorem 1(c)). The certificate $\bar{h}(T_t \mid t) - \frac{1}{2}\ln(2\pi e\,\mathrm{MSE})$ is a lower bound on $I(T_t; H_{1:t} \mid t)$. A predictor that uses only the age is a negative control, since it cannot exceed the bound; its certificate is -0.11, -0.11, -0.22 and -0.12 nats on FD001 to FD004. The HI trajectory certifies information beyond the age (Figure 6(a)): the ridge HI certifies 0.31, 0.21, 0.10 and 0.18 nats, with 95% intervals excluding zero except on FD003. Level-only and slope-only summaries of the same HI certify less than the full trajectory on FD001, FD002 and FD004 (0.07 to 0.17 nats on FD001), and a predictor on the raw sensors certifies 0.16 to 0.25 nats, similar to the supervised HI (and higher on FD003 and FD004), so on these data the HI is a compression of the sensors that retains the certified information at best, not an addition to it. With 100 or more engines the certificate is therefore non-vacuous and passes its control.

Lithium-ion cells. The age-only control is at or below zero (-0.02 nats on HUST, -0.27 on MATR and 0.05 on Tongji, whose interval includes zero), and the ridge HI certifies 0.30, 0.32 and 0.60 nats with intervals above zero (Figure 6(a), Table F.21). The three charging features used directly certify more (0.67, 0.61 and 0.91 nats) than the scalar HI constructed from them, as the data-processing inequality requires, and the capacity reference does not exceed them (0.59, 0.29 and 0.40 nats).

Small data sets. On the seven small battery and bearing sets of Section 6 the model-based age-conditional certificate fails the negative control on four of them, because a log-normal lifetime law fitted to three to seven lifetimes overestimates $h(T_t \mid t)$ (Tables D.15 and D.16 in Appendix D), which is why this test needs many units; it was informative and passed its control only with 77 or more.

5.3. Squeezing

Turbofan engines (Theorem 3). A squeeze toward the mean HI trajectory of the training engines, with noise chosen as in Section 4.1, raises the monotonicity of the ridge HI from 0.13–0.24 to 0.41–0.58 (Table F.22), where it saturates for $\gamma \geq 0.97$ at the value predicted

from the template (0.55, 0.58, 0.42, 0.41 observed against 0.55, 0.61, 0.41, 0.41 predicted for FD001 to FD004; Figure 6(b)). Over the same range the certified information falls from 0.10–0.31 nats to about zero or below (Figure 6(c)).

Lithium-ion cells. Squeezing raises the monotonicity of the ridge HI from 0.32 to 0.81 on HUST, from 0.39 to 0.48 on MATR and from 0.65 to 0.73 on Tongji, close to the predicted limits (0.82, 0.49 and 0.70; the limit is 0.03 too low on Tongji), while the certificate falls from 0.30 to 0.00, from 0.32 to -0.20 and from 0.62 to 0.00 nats (Figure 6(b,c)). In both domains, monotonicity and certified information are decoupled.

5.4. Direction of a scalar linear HI

Turbofan engines (Theorem 4, Corollary 1). For the level and the velocity of the ridge HI, the direction of the best linear scalar combination of the RUL moves across stages (for example between 21° and 56° across age stages on FD002, Table F.23), and level-only and velocity-only HIs lose up to 20% and 38% of RMSE relative to the free direction (Figure 7(b,c)). However, the best *static* direction, combined with a stage-specific affine map, loses at most 4% in every stage and subset. The law of Section 4.2 is therefore consistent with the observed rotation, but its consequence is small on these data: it is material only when the angle between the stage optima is large relative to the signal-to-noise ratio, which is not the case here.

Lithium-ion cells. On HUST the optimal direction moves from 71°–80° in the first three life stages to 3°–16° in the last two, and the best static direction loses up to 9% of RMSE in a stage (18% in the last age stage), against up to 54% for velocity alone (Figure 7(b,c), Table F.23). On MATR the optimal direction also rotates across the life fraction (88° to 42°), but a static direction loses at most 1.2%; on Tongji it loses at most 3% across life stages and 7% across age stages, against up to 35% for the level alone. The loss predicted by the angular law is thus material for HUST, modest for Tongji and negligible for MATR and for engines.

5.5. Regime and protocol leakage

Turbofan engines (Proposition 1). For a linear read-out of the life fraction, regime blindness increases the RMSE from 0.137 to 0.148 on FD002 and from 0.165 to 0.199 on FD004 (8% and 21%); with one regime the three read-outs coincide (Table F.24). Per-regime normalisation removes most of the loss, and regime-specific weights add only 1%, so the residual effect of heterogeneous gains is small in these data.

Lithium-ion cells. A linear read-out of the life fraction from the raw charging features, standardised over the population, has an RMSE of 0.136 on HUST and 0.179 on MATR; expressing each feature relative to the own early cycles of the cell, which removes the protocol- and cell-specific level, lowers it to 0.115 and 0.142 (18% and 26%; Table F.24), and from 0.190 to 0.167 (14%) on Tongji. This is the analogue of regime normalisation of Proposition 1, with the cell playing the role of the regime.

6. Benchmark of estimators

Section 5 tests the theory with generic HIs. This section asks what the theory implies for estimators, and makes three predictions that run-to-failure data can test. (i) A

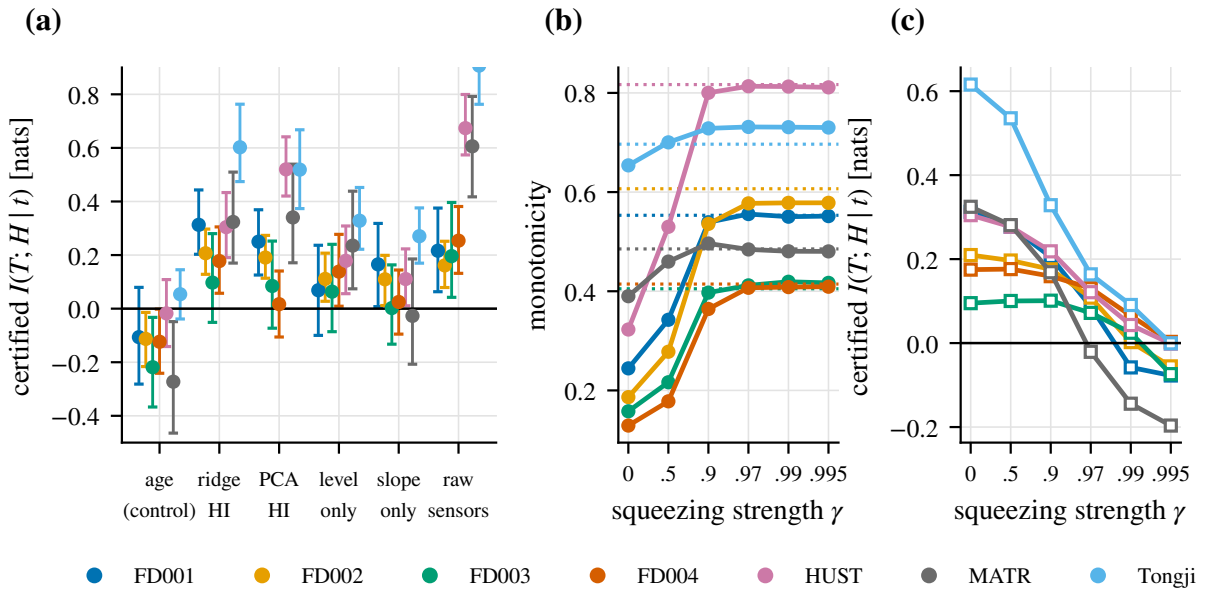


Figure 6: Cross-validated tests on turbofan engines (FD001–FD004) and lithium-ion cells (HUST, MATR, Tongji). (a) Certified $I(T_t; H_{1:t} \mid t)$ of HI summaries, with 95% bootstrap intervals. (b) Monotonicity and (c) certified information after a squeeze of strength γ; dotted lines: predicted Mon_∞.

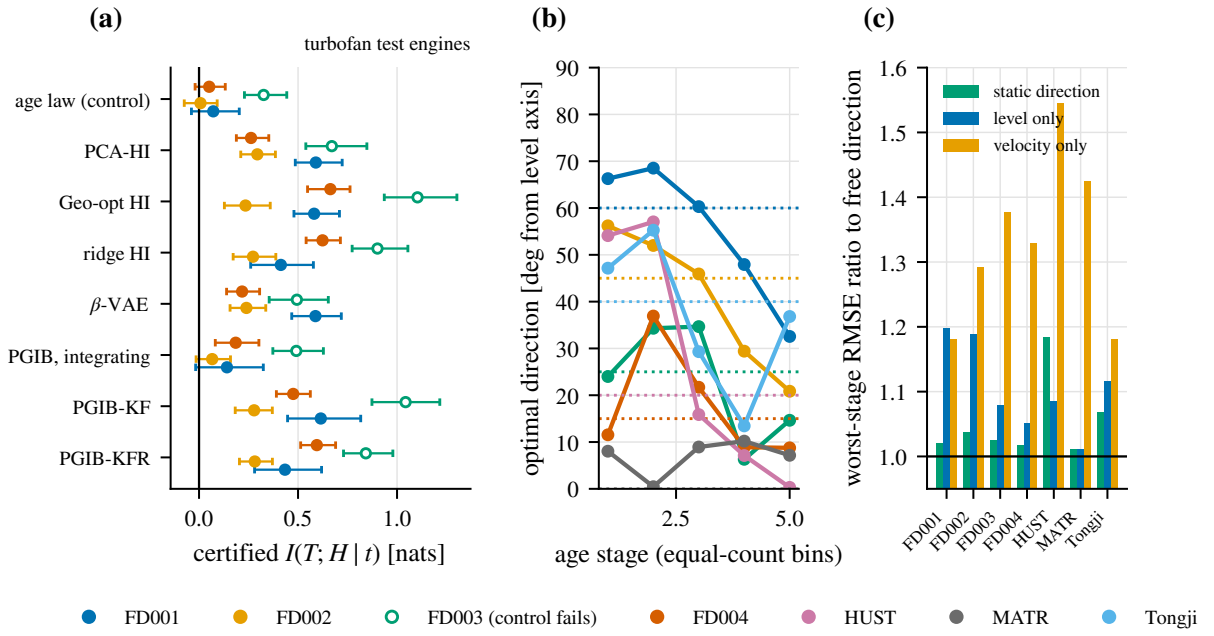


Figure 7: (a) Certificates of the benchmark methods on the official turbofan test engines (hollow markers: age-only control fails). (b) Optimal direction of a scalar linear HI per age stage; dotted: best static direction. (c) Worst-stage RMSE ratio to the free direction.

sensor-free predictor that uses only the age should be competitive when the sensors are weak (Theorem 1(c)). (ii) An integrating encoder accumulates error and an observation-corrected one does not (Proposition 2). (iii) Learned encoders have a training variance that must be measured before differences between HIs are interpreted.

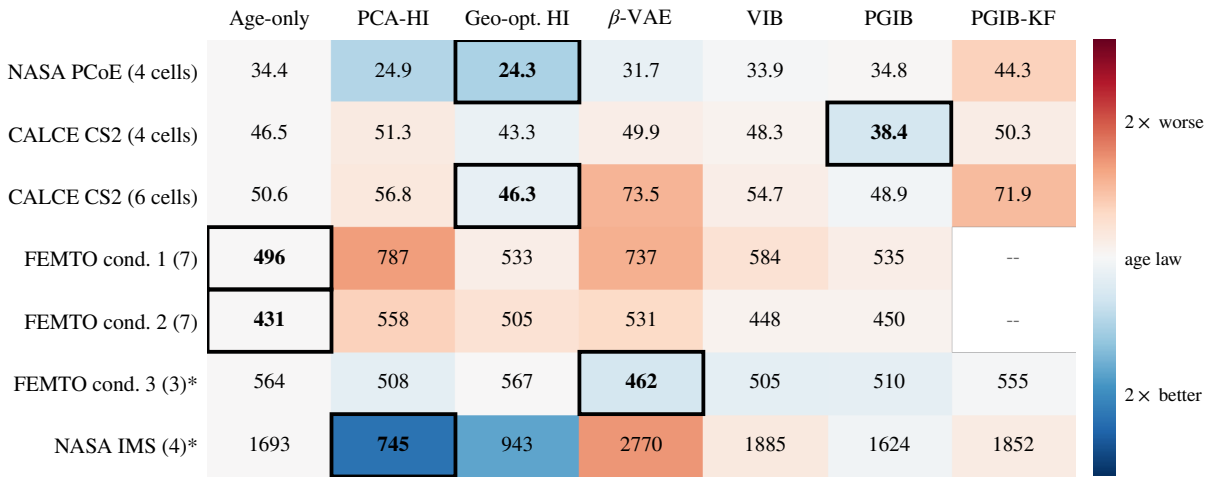


Figure 8: RMSE of the compared HIs on the seven small data sets (cycles for batteries, 10 s snapshots for bearings). Colour: ratio to the age law; frame: lowest value of each row; *: sets not used in the design.

Design. The data are battery cells from the NASA Prognostics Center of Excellence (PCoE) and from the Center for Advanced Life Cycle Engineering (CALCE, CS2 cells, in sets of four and six); bearings from the FEMTO-ST institute (PRONOSTIA, conditions 1 to 3) and from the NASA Center for Intelligent Maintenance Systems (IMS); and the four C-MAPSS subsets FD001 to FD004, of which FD001 and FD003 have one operating regime and FD002 and FD004 six. Three simple HIs are compared with three learned encoders. The simple HIs are the first principal component (PCA-HI), a linear combination optimised for the sum of the geometric criteria monotonicity, trendability and prognosability (Geo-opt HI), and a supervised ridge regression (ridge HI). The learned encoders are a β-variational autoencoder (β-VAE), a variational information bottleneck without physics prior (VIB), and the PGIB of Appendix B with its Kalman-corrected (PGIB-KF) and regime-aware (PGIB-KFR) variants. All HIs feed the *same* calibrated predictor of Appendix B.4. Evaluation is leave-one-unit-out for the small sets and uses the official test engines for C-MAPSS. Data, protocols, their history and the evidence tiers (which results are clean confirmations and which are development results) are given in Appendix C, and full tables in Appendix D.

Age and few training units (prediction i). With two to six training units no method is best everywhere (Figure 8): the age law has the lowest RMSE on both FEMTO conditions and is within a factor of 1.5 of the best method on six of the seven sets. On C-MAPSS the sensors are far more informative: the age law has an RMSE of 36.6 to 54.7 cycles, against 21.8 to 29.7 for the best simple HI. The age law is thus a strong baseline where the sensors are weak, and it must be reported.

Error accumulation and its repair (prediction ii). With 80 to 208 training engines, the integrating encoder is well behind the best simple HI (single-model RMSE 34.8 to 48.7 against 21.8 to 29.7) and only 2 to 6 RMSE units below the age law. The observation-corrected PGIB-KF reduces the RMSE to 23.8–40.7 for a single model and to 21.3–35.8 for a ten-seed ensemble (Table 4). The deficit is therefore not caused by the bottleneck objective, which is the same in both models, but by the missing anchoring of the level.

Table 4: C-MAPSS, ten seeds per learned encoder. Single: RMSE, mean ± standard deviation over seeds; ens. 10: ten-seed ensemble; last column: paired difference of PGIB-KF from the best simple HI with 95% bootstrap interval over engines [66].

Subset	Best simple HI (RMSE)	PGIB-HI single	PGIB-KF-HI single	PGIB-KFR-HI single	PGIB-KF-HI ens. 10	PGIB-KFR-HI ens. 10	KF − best paired, ens. 10 [95% CI]	KFR − best
FD001	PCA (21.8)	34.8±2.9	23.8±3.3	27.5±3.2	21.3	25.5	-0.6 [-3.2, +2.5]	+3.6 [+0.2, +7.0]
FD002	regime ridge (28.4)	36.3±1.5	31.0±2.2	30.6±2.2	29.2	29.1	+0.8 [-0.7, +2.3]	+0.7 [-1.2, +2.8]
FD003	Geo-opt. (21.9)	41.2±1.9	30.1±4.6	33.8±6.5	23.3	28.4	+1.4 [-1.6, +4.5]	+6.5 [+2.4, +10.6]
FD004	Geo-opt. (29.7)	48.7±2.5	40.7±7.6	36.8±4.0	35.8	31.8	+6.1 [+2.5, +9.7]	+2.1 [-0.9, +5.0]

Certificates of the benchmark methods. Using the lifetime law of Section 5, the certificates of the methods compared here on the official test engines (Figure 7(a), Table F.25) provide an information-theoretic reading of the error accumulation of Proposition 2. On FD001, FD002 and FD004 the age law passes the control (0.07, 0.01, 0.05 nats). The integrating PGIB certifies only 0.14, 0.07 and 0.19 nats, which is close to the age law on FD001 and FD002, whereas the observation-corrected PGIB-KF certifies 0.61, 0.28 and 0.48 nats and the best simple HI 0.59, 0.29 and 0.66 nats. On FD003 the control fails (age law: 0.33 nats, interval 0.23 to 0.44), so the lifetime law of the training engines does not describe the test population there and the FD003 values are not interpreted.

Training variance (prediction iii). The RMSE of a single learned encoder has a standard deviation of 1.5 to 7.6 units over seeds, as much as the differences between competing HIs (Figure 9), which calls for variance-aware comparisons [65]. A three-seed ensemble on FD003 suggested that PGIB-KF beats the best simple HI (19.4 against 21.9), but with ten seeds [38] the difference disappears (23.3; paired difference +1.4, interval −1.6 to 4.5). With ten seeds, the PGIB-KF ensemble is statistically tied with the best simple or supervised HI on FD001 to FD003, worse on FD004, and never significantly better. The regime-aware variant is not shown to help: it is worse than PGIB-KF on the single-regime subsets FD001 and FD003 (single-model RMSE 27.5 against 23.8 and 33.8 against 30.1) and better only on FD004, where the difference is not significant.

Role of the age fusion. The age fusion lowers the RMSE of most HIs considerably (for example PCA-HI on CALCE with four cells from 186 to 51 cycles) but not always (NASA), and it narrows the differences between HIs. Without fusion, that is, with a predictor based on the HI alone, the PGIB has the lowest RMSE on CALCE (four and six cells) and FEMTO condition 2, VIB on FEMTO condition 1, the β-VAE on FEMTO condition 3 and PCA-HI on NASA and IMS (Table D.13). The conclusion that no learned encoder is superior therefore holds for the common age-fused predictor and is more mixed for HI-only predictors, for which the information in the HI is less masked by the age prior.

7. Engineering guidance

This section converts the results into decisions. Table 5 lists the question an engineer faces, the recommendation that follows from the theory, and the size of the effect observed in Sections 5 and 6.

Workflow. The recommendations combine into four steps. (1) Fit the age-only baseline and compute its certificate, which must not exceed zero. (2) Build candidate HIs from sensors normalised by regime or by the unit's early cycles, and supply their level and velocity to the predictor. (3) Compute the age-conditional certificate of each candidate

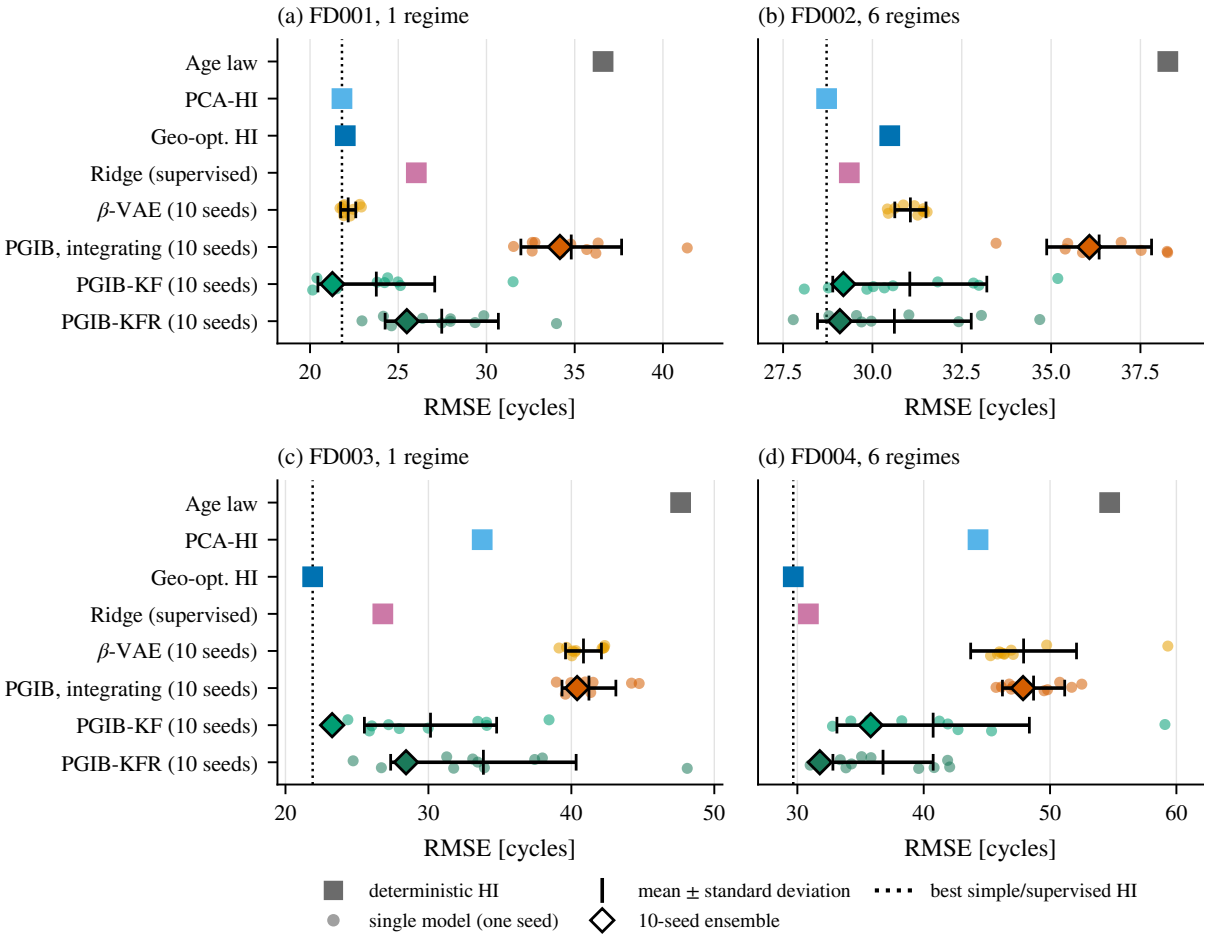


Figure 9: C-MAPSS test RMSE. Squares: deterministic HIs; dots: single models of each learned encoder; bars: mean and standard deviation; diamonds: ten-seed ensembles; dotted line: best simple or supervised HI.

by unit-grouped cross-validation with bootstrap intervals over units, and keep the HI with the highest certified information whose interval excludes zero. (4) Compare the achieved RMSE with the aleatoric floor to judge how much a better HI could still gain. The certificate was informative with 77 or more units and failed its control on four of the seven sets with three to seven units; we have no result in between, and with few units the age baseline and the simplest HI are the safe choice.

8. Limitations and conclusions

Limitations.

1. *Theory.* The theory of Section 4.2 is local and Gaussian, Theorem 2 assumes a constant-drift Wiener future, Proposition 1 concerns a scalar sensor and a regime-blind linear read-out, and Theorem 4 concerns scalar linear HIs with fixed weights. Non-Gaussian posteriors, non-constant drift, regime switching, nonlinear multi-sensor estimators and the extension to snapshot-level projection limits [54] are open, and the entropy estimates in the simulation checks have unquantified Monte Carlo error.

Table 5: Engineering guidance: decisions in HI design, recommendations and the effects observed.

Question	Recommendation	Effect observed
Does my HI keep the information a predictor needs?	Compute the age-conditional certificate (Theorem 1(c)) by unit-grouped cross-validation, with the age-only predictor as negative control, and compare HIs in nats; 0.1 nat lowers the bound on the RMSE by about 10%.	The ridge HI certifies 0.10 to 0.60 nats on engines and cells while the control stays at or below 0.05 nats; on four of the seven sets with three to seven units the control fails.
Which HI should I keep?	Do not select or tune HIs by monotonicity, trendability or prognosability; rank them by certified information.	Squeezing raised the monotonicity from 0.13–0.24 to 0.41–0.58 on engines and from 0.32 to 0.81 on HUST cells while the certified information fell to about zero.
One scalar, or level and velocity?	Give the predictor the level and the velocity of the HI, or let the weights follow the stage of life.	A static direction costs at most 4% RMSE on engines, 1.2% on MATR and 7% on Tongji cells, but 9 to 18% on HUST cells; level alone loses up to 35% and velocity alone up to 54%.
Do operating regimes or test protocols matter?	Normalise the sensors by regime and, when protocols differ, by each unit's own early cycles; give the regime to the observation model when gains differ.	Without normalisation the RMSE of a linear read-out is higher by 8% (FD002), 21% (FD004), 18% (HUST), 26% (MATR) and 14% (Tongji).
Should an encoder integrate the velocity?	Not without an observation correction, that is, a learned gain on a level read-out.	On C-MAPSS the integrating encoder has an RMSE of 34.8 to 48.7 and the corrected one 23.8 to 40.7 (single model) and 21.3 to 35.8 (ten-seed ensemble).
Is a learned encoder worth its cost?	Start from a simple HI with the same age-fused predictor and adopt an encoder only if its certified information is higher.	No learned encoder beat the best simple HI; ten trainings of a recurrent model were needed to tie it on C-MAPSS, and the age law was within a factor of 1.5 of the best method on six of seven small sets.
How much can a better HI still gain?	Compare the achieved error with the aleatoric floor of Theorem 2.	In the simulated example 42% of the squared error of a lossless HI is irreducible.

2. *Certificates.* The efficiency η_{info} could not be estimated reliably on real data. The certificate is a lower bound that depends on the predictor and, in its age-conditional form, on a lifetime law that must be validated by a negative control, which failed for the test population of FD003. It is vacuous for most small data sets, where the information that an HI adds to the age could not be certified (Section 5).
3. *Benchmark design.* The regime-aware variant and the variance study were designed after seeing results on the sets used, and no independent multi-regime data set was available (evidence tiers in Appendix C). Ensembles of ten encoders are compared with deterministic simple HIs, which favours the encoders in computation. The age fusion and the out-of-sample calibration were introduced after an initial protocol showed under-coverage; they benefit most HIs but are a pragmatic device, not a Bayes-exact fusion, and they mask part of the differences between HIs.
4. *Data.* The bearing features are generic time-domain and spectral summaries, and bearing degradation is heterogeneous; physics-based features may change the picture. The battery and bearing sets of the benchmark have three to seven units, and the benchmark on C-MAPSS is evaluated at the last observed cycle of each test en-

gine, not along the test trajectories. The real-data tests use cross-validation within the data sets, cover simulated turbofan engines and three collections of lithium-ion cells (lithium iron phosphate and three nickel-based chemistries) with charging-phase features, and include no bearing data set with more than a handful of units because, to our knowledge, no public run-to-failure bearing collection of measured data has more than 17 bearings (FEMTO-ST and the Paderborn time-varying set have 17, XJTU-SY has 15; a survey of 110 public data sets lists seven for bearing prognosis [67]).

Conclusions. For any HI, an engineer can ask how much information about the RUL it keeps beyond the age, and the answer, in nats, follows from the held-out error of any predictor. The theory identifies four ways in which an HI loses information, and the tests on turbofan engines and lithium-ion cells show that they matter in practice: monotonicity scores can rise while information vanishes, a static scalar HI costs up to 18% RMSE in one cell collection, ignoring the operating regime or protocol costs 8 to 26%, and an integrating encoder needs an observation correction. With few units the age law is a strong baseline, and no learned encoder beat simple indicators. The contribution of the paper is the budget and the decisions it supports, not a new predictor.

CRediT authorship contribution statement

Withheld for anonymous review.

Declaration of competing interest

Withheld for anonymous review.

Declaration of generative AI and AI-assisted technologies in the writing process

Withheld for anonymous review.

Funding

Withheld for anonymous review.

Data availability

NASA PCoE, NASA IMS, C-MAPSS, CALCE CS2, FEMTO-ST/PRONOSTIA, HUST, MATR and Tongji data are publicly available (NASA Prognostics Data Repository, CALCE Battery Group, IEEE 2012 Prognostics and Health Management data challenge, BatteryLife repository on Zenodo). The simulation and benchmark code, the stored results and the per-unit outputs used to generate all tables and figures are available in an anonymised repository: `https://anonymous.4open.science/r/information-budget-health-indicator-rul-C5CE/`.

Acknowledgements

Withheld for anonymous review.

A. Proofs and supplementary results

A.1. Relation to the Bayesian Cramér–Rao bound

Remark 3 (Relation to the Bayesian Cramér–Rao bound)**.** *Let $J(h)$ be the Fisher information, with respect to location, of the posterior density $p(T_t \mid H_{1:t} = h)$. Under the regularity conditions of the van Trees inequality [51, 52] every estimator satisfies $\mathbb{E}[(T_t - \hat{T}_t)^2] \geq 1/\mathbb{E}_H[J(H)]$. Stam's inequality [68] states that $e^{2h(T_t|H=h)}/(2\pi e) \geq 1/J(h)$ for every h. Taking the geometric mean over h and using that the geometric mean of $1/J$ is at least the reciprocal of the arithmetic mean of J,*

$$\mathcal{B}_{\text{info}}(H_{1:t}) = \exp\Big(\mathbb{E}_H \ln \frac{e^{2h(T_t|H=h)}}{2\pi e}\Big) \;\geq\; \exp\Big(\mathbb{E}_H \ln \frac{1}{J(H)}\Big) \;\geq\; \frac{1}{\mathbb{E}_H[J(H)]}, \tag{A.1}$$

so, wherever the Bayesian Cramér–Rao bound applies, the entropy-power bound is not weaker, and it requires no differentiability of the densities. For posterior filtering the Fisher-information route is developed in [53].

A.2. State–kinematic decomposition and pointwise squeezing

Proposition 3 (State–kinematic decomposition)**.** *Let $\Delta H_k = H_k - H_{k-1}$ denote the increments of the HI. The map $H_{1:t} \mapsto (H_t, \Delta H_{2:t})$ is a linear bijection, hence*

$$I(T_t; H_{1:t}) = \underbrace{I(T_t; H_t)}_{\mathcal{I}_{\text{state}}(t)} + \underbrace{I(T_t; \Delta H_{2:t} \mid H_t)}_{\mathcal{I}_{\text{kin}}(t)}. \tag{A.2}$$

Proof. The map has a unit-determinant matrix, so $h(\mathbf{AY}) = h(\mathbf{Y}) + \ln|\det \mathbf{A}|$ [40, Sect. 8.6] and mutual information is unchanged ($H_k = H_t - \sum_{j=k+1}^{t} \Delta H_j$ gives the inverse). The identity is then the chain rule of mutual information [40, Sect. 2.5]. □

Remark 4. *The decomposition is an identity; its content lies in which operations remove which part. If the drift is constant, H_t/t is a sufficient statistic for the drift, and $\mathcal{I}_{\text{kin}}$ about the drift is zero. Kinematic information is non-trivial when the velocity varies in time, for example under a state-dependent drift or a random-walk velocity, as in late-life acceleration.*

Proposition 4 (Pointwise squeezing)**.** *Let the squeeze of Theorem 3 act pointwise in time, $\tilde{H}_k = \phi_k(H_k, \xi_k)$. Then*

$$\begin{gathered} \ln \frac{\mathcal{B}_{\text{info}}(\tilde{H}_{1:t})}{\mathcal{B}_{\text{info}}(H_{1:t})} = 2\big(\Delta\mathcal{I}_{\text{state}} + \Delta\mathcal{I}_{\text{kin}}\big), \\ \Delta\mathcal{I}_{\text{state}} := \mathcal{I}_{\text{state}}(H) - \mathcal{I}_{\text{state}}(\tilde{H}) \geq 0, \quad \Delta\mathcal{I}_{\text{kin}} := \mathcal{I}_{\text{kin}}(H) - \mathcal{I}_{\text{kin}}(\tilde{H}), \end{gathered} \tag{A.3}$$

where $\Delta\mathcal{I}_{\text{kin}}$ has no definite sign in general, so that kinematic information alone does not determine the inflation. In the Gaussian squeeze of Theorem 3(b), write $\mathcal{I}^{\vartheta}_{\text{state}}(\gamma) = I(\vartheta; \tilde{H}^{\gamma}_t)$ and $\mathcal{I}^{\vartheta}_{\text{kin}}(\gamma) = I(\vartheta; \Delta\tilde{H}^{\gamma}_{2:t} \mid \tilde{H}^{\gamma}_t)$; then $\mathcal{I}^{\vartheta}_{\text{state}}$ is non-increasing in γ and $0 \leq \mathcal{I}^{\vartheta}_{\text{state}}, \mathcal{I}^{\vartheta}_{\text{kin}} \leq \mathcal{I}^{\vartheta}_{\text{tot}}(\gamma) \to 0$, although $\mathcal{I}^{\vartheta}_{\text{kin}}$ need not be monotone.

Proof. Apply Proposition 3 to H and to $\tilde{H}$ and take the difference in Theorem 3(a). Because $\tilde{H}_t = \phi_t(H_t, \xi_t)$ is a randomised function of H_t alone, the DPI gives $I(T_t; \tilde{H}_t) \leq I(T_t; H_t)$, that is, $\Delta\mathcal{I}_{\text{state}} \geq 0$. For non-pointwise operators (for example smoothing over history) even this sign can fail. In the Gaussian squeeze, the argument of the proof of Theorem 3(b) applied to the scalar observation at time t gives the monotonicity of $\mathcal{I}^{\vartheta}_{\text{state}}$; since $\mathcal{I}^{\vartheta}_{\text{state}} = I(\vartheta; \tilde{H}_t) \leq I(\vartheta; \tilde{H}_{1:t})$ and $\mathcal{I}^{\vartheta}_{\text{kin}} = \mathcal{I}^{\vartheta}_{\text{tot}} - \mathcal{I}^{\vartheta}_{\text{state}} \leq \mathcal{I}^{\vartheta}_{\text{tot}}$, both are dominated by $\mathcal{I}^{\vartheta}_{\text{tot}}$, which vanishes as $\gamma \to 1$. □

A.3. Proof of Theorem 3

Proof. (a) $\tilde{H}$ is a randomised function of H only, so the Markov property holds and the DPI gives the first claim; the ratio follows from the definition of $\mathcal{B}_{\text{info}}$. (b) With $\lambda = 1-\gamma > 0$, $\tilde{H}^\gamma$ is an invertible affine image of $(H-m)+\xi^{(2)}/\lambda$, that is, of H observed through additive Gaussian noise of variance σ_2^2/λ^2. For $\gamma_2 > \gamma_1$ the noise of the second channel is the first plus independent Gaussian noise, so the DPI [40, Sect. 2.8] gives monotonicity of $\mathcal{I}_{\text{tot}}^{\vartheta}$. As $\lambda \to 0$ the noise variance diverges, and the mutual information of a Gaussian channel, $\frac{1}{2}\ln(1+\text{SNR})$ per scalar observation with signal-to-noise ratio SNR [40, Chap. 9], vanishes. For the limit of Mon, as $\lambda \to 0$ the increments converge to $\Delta m + \xi_{k+1}^{(2)} - \xi_k^{(2)}$, a stationary 1-dependent sequence. The indicators of a positive increment at odd k involve disjoint pairs of the $\xi^{(2)}$ and are therefore i.i.d., as are those at even k, so the strong law of large numbers [69, Chap. 2] applies to each subsequence and the fraction of positive increments tends to $p = \Phi_{\mathcal{N}}(\Delta m/(\sqrt{2}\sigma_2))$ and Mon $\to |2p-1|$. □

A.4. Non-Gaussianity gap

Proposition 5 (Non-Gaussianity gap)**.** *Let $\hat{m}_H = \mathbb{E}[m \mid H]$ be the Bayes-optimal estimator of m and let $D_h := D_{\text{KL}}\big(p(m \mid H=h) \,\|\, \mathcal{N}(\mathbb{E}[m \mid h], \text{Var}(m \mid h))\big) \geq 0$ be the Kullback–Leibler (KL) divergence of the posterior from the Gaussian with the same mean and variance. Then*

$$\mathbb{E}\big[(m-\hat{m}_H)^2\big] - \mathbb{E}_H\left[\frac{e^{2h(m|H=h)}}{2\pi e}\right] = \mathbb{E}_H\Big[\,\text{Var}(m \mid H)\big(1-e^{-2D_H}\big)\Big] \;\geq 0, \tag{A.4}$$

with equality if and only if $p(m \mid H)$ is Gaussian almost surely.

Proof. For a density p with mean μ_p and variance σ_p^2 and the Gaussian $q = \mathcal{N}(\mu_p, \sigma_p^2)$, $\int p \ln q = -\frac{1}{2}\ln(2\pi\sigma_p^2) - \frac{1}{2}$ because q depends on x only through the first two moments of p. Hence $D_{\text{KL}}(p\|q) = -h(p) + \frac{1}{2}\ln(2\pi e\sigma_p^2)$ (the identity behind the maximum-entropy property of the Gaussian [40, Sect. 8.6]), that is, $e^{2h(p)}/(2\pi e) = \sigma_p^2 e^{-2D_{\text{KL}}(p\|q)}$. Apply this to $p = p(m \mid H=h)$, use $\mathbb{E}[(m-\hat{m}_H)^2 \mid H=h] = \text{Var}(m \mid h)$, which holds because the posterior mean is the Bayes-optimal estimator under squared loss [57, Sect. 1.5.5], and average over h. The right-hand side vanishes if and only if $D_H = 0$ almost surely, that is, if and only if the posterior is Gaussian. □

A.5. Reduction to functionals of the sufficient statistic

Lemma 1 (Reduction to functionals of $\hat{\mathbf{z}}$)**.** *For every scalar linear functional $w^\top \mathbf{X}_{1:t}$ there is $\mathbf{u}$ such that* $\text{corr}(m_L, w^\top\mathbf{X})^2 \leq \text{corr}(m_L, \mathbf{u}^\top\hat{\mathbf{z}})^2$*. Hence no scalar linear HI is more informative about m_L than the best $h_{\mathbf{u}}$. Here* corr *denotes the correlation coefficient.*

Proof. Let $\mathbf{C} = \text{Cov}(\mathbf{z}, \mathbf{X})$ and $\mathbf{V} = \text{Var}(\mathbf{X})$, so $\hat{\mathbf{z}} = \mathbf{z}_0 + \mathbf{C}\mathbf{V}^{-1}(\mathbf{X} - \mathbb{E}\mathbf{X})$ and $\mathbf{\Lambda} = \mathbf{C}\mathbf{V}^{-1}\mathbf{C}^\top$. Put $\mathbf{u} = \mathbf{\Lambda}^{-1}\mathbf{C}w$. Then $\text{Cov}(\hat{\mathbf{z}}, w^\top\mathbf{X}) = \mathbf{C}\mathbf{V}^{-1}\mathbf{V}w = \mathbf{C}w = \mathbf{\Lambda}\mathbf{u}$, so $w^\top\mathbf{X} = \mathbf{u}^\top\hat{\mathbf{z}} + e$ with e uncorrelated with $\hat{\mathbf{z}}$ and, since $\text{Cov}(\mathbf{z}, w^\top\mathbf{X}) = \mathbf{C}w = \mathbf{\Lambda}\mathbf{u} = \text{Cov}(\mathbf{z}, \mathbf{u}^\top\hat{\mathbf{z}})$, also uncorrelated with $\mathbf{z}$. Therefore $\text{Cov}(m_L, w^\top\mathbf{X}) = \text{Cov}(m_L, \mathbf{u}^\top\hat{\mathbf{z}}) = \mathbf{g}^\top\mathbf{\Lambda}\mathbf{u}$ while $\text{Var}(w^\top\mathbf{X}) = \mathbf{u}^\top\mathbf{\Lambda}\mathbf{u} + \text{Var}(e) \geq \text{Var}(\mathbf{u}^\top\hat{\mathbf{z}})$, which gives the claim. □

A.6. Proof of Corollary 1

Proof. (a) By Theorem 4, the gradient of the frozen-velocity RUL at the operating point is $\mathbf{g} = -\bar{V}^{-1}(1, \bar{a}/\bar{V})^\top$, which is proportional to $(1, \bar{\tau})^\top$; the Euclidean angle of this direction from the level axis is $\arctan \bar{\tau}$. (b) By Theorem 4(c), inflation at stage j vanishes only for $\mathbf{u} \parallel \mathbf{g}_j$, so it vanishes at both stages only if $\mathbf{g}_1 \parallel \mathbf{g}_2$. The map $\mathbf{u} \mapsto \max_j \rho_j \sin^2 \psi_j(\mathbf{u})$ is continuous and positive on the unit circle, hence has a positive minimum δ. For the quantitative statement note that the $\mathbf{\Lambda}$-angle between lines satisfies the triangle inequality $\psi_{12} \le \psi_1 + \psi_2$. Indeed, for unit vectors a, b in the $\mathbf{\Lambda}$-inner product the angle $\arccos\langle a, b\rangle$ is the geodesic distance on the unit sphere, which is a metric, and the angle between the lines through a and b is $\min_{s=\pm 1} \arccos\langle a, sb\rangle$. Choose the sign of the representative of $\mathbf{u}$ so that its angle to the representative of $\mathbf{g}_1$ is ψ_1, then the sign of the representative of $\mathbf{g}_2$ so that its angle to that of $\mathbf{u}$ is ψ_2; the spherical triangle inequality bounds the angle between the representatives of $\mathbf{g}_1$ and $\mathbf{g}_2$ by $\psi_1 + \psi_2$, and ψ_{12} is not larger. Therefore $\max_j \psi_j \ge \psi_{12}/2$ and $\sin^2$ is increasing on $[0, \pi/2]$. The last sentence is Theorem 4(c) applied at each stage. □

B. Estimators used in the benchmark

This appendix describes the estimators compared in Section 6. The path-space objective follows from the theory; the architectures and the calibration are design choices.

B.1. Path-space objective

We seek a causal stochastic encoder, the *physics-guided information bottleneck* (PGIB), $q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) = \prod_{k=1}^{t} q_\theta(H_k \mid H_{k-1}, \mathbf{X}_{1:k})$, whose increments are Gaussian, $\mathcal{N}\big(H_{k-1} + u_\theta(\mathbf{X}_{1:k})\Delta t,\ s_\theta^2(\mathbf{X}_{1:k})\Delta t\big)$, with a positive velocity u_θ (enforced by a softplus output) and a variance s_θ^2. The encoder is regularised towards a physics prior $p_\vartheta(H_{1:t}) = \prod_k \mathcal{N}\big(H_{k-1} + \eta(H_{k-1}; \vartheta)\Delta t,\ \sigma_B^2 \Delta t\big)$, $H_0 \equiv 0$, where Δt is the time step, η a state-dependent drift with parameters ϑ, and σ_B the diffusion coefficient.

Proposition 6 (Compression bound and path-space KL)**.** *For any prior p_ϑ, $I(\mathbf{X}_{1:t}; H_{1:t}) \le \mathbb{E}_{\mathbf{X}}\big[D_{\mathrm{KL}}(q_\theta(H_{1:t} \mid \mathbf{X}_{1:t}) \,\|\, p_\vartheta(H_{1:t}))\big]$, and for the Gaussian factorisation above the divergence is exactly*

$$D_{\mathrm{KL}} = \frac{1}{2} \sum_{k=1}^{t} \mathbb{E}_{q_\theta(H_{k-1} \mid \mathbf{X}_{1:k-1})} \left[\frac{s_\theta^2}{\sigma_B^2} - 1 - \ln \frac{s_\theta^2}{\sigma_B^2} + \frac{(u_\theta - \eta(H_{k-1}; \vartheta))^2 \, \Delta t}{\sigma_B^2} \right]. \tag{B.1}$$

Proof. The first claim is the standard variational upper bound of the information bottleneck [41–43]: $I(\mathbf{X}; H) = \mathbb{E}_{\mathbf{X}} D_{\mathrm{KL}}(q(H \mid \mathbf{X}) \| q(H)) \le \mathbb{E}_{\mathbf{X}} D_{\mathrm{KL}}(q(H \mid \mathbf{X}) \| p(H))$ because the difference is $D_{\mathrm{KL}}(q(H) \| p(H)) \ge 0$. The chain rule of the KL divergence [40, Sect. 2.5] over the causal factorisation gives $\sum_k \mathbb{E}_{q(H_{k-1} \mid \cdot)} D_{\mathrm{KL}}(q(H_k \mid H_{k-1}, \mathbf{X}_{1:k}) \| p_\vartheta(H_k \mid H_{k-1}))$, and the Gaussian KL with means differing by $(u_\theta - \eta)\Delta t$ and variances $s_\theta^2 \Delta t$, $\sigma_B^2 \Delta t$ yields the summand (the Gaussian KL of the variational autoencoder [20]). □

The training objective is $\mathcal{L}_{\mathrm{PGIB}} = \mathcal{L}_{\mathrm{FHT}} + \beta \, D_{\mathrm{KL}}$, where β weights the compression term and $\mathcal{L}_{\mathrm{FHT}}$ is the negative log-likelihood of the observed failure time under the predictive law below, a variational surrogate for $-I(T_t; H_{1:t})$ in the sense of [42] (Algorithm 1 in Appendix C).

B.2. Predictive law

Let the posterior of the unit's drift μ given the HI increments be $\mathcal{N}(\hat{\mu}_t, \sigma^2_{\mu,t})$ (conjugate update). Marginalising the inverse-Gaussian law over μ (a Gaussian integral; random-drift Wiener models are treated in [28, 29]) gives

$$p(T_t = \tau \mid H_{1:t}) = \frac{a}{\sqrt{2\pi\tau^3(\sigma_B^2 + \sigma^2_{\mu,t}\tau)}} \exp\!\left(-\frac{(a - \hat{\mu}_t\tau)^2}{2\tau(\sigma_B^2 + \sigma^2_{\mu,t}\tau)}\right), \qquad a = D_{\text{fail}} - \hat{D}_t. \tag{B.2}$$

Here σ_B^2 is the aleatoric and $\sigma^2_{\mu,t}\tau$ the epistemic contribution to the effective diffusion; as t grows $\sigma^2_{\mu,t} \to 0$ and (B.2) reduces to the inverse-Gaussian law. It is a defective density, the missing mass being the probability of a non-positive drift, and $\hat{D}_t$ is the filtered state rather than the raw HI value, which avoids the degeneracies of non-monotone paths. The law was checked against numerical integration of the mixture (Section 3).

B.3. Observation-corrected and regime-aware encoders

The encoder above integrates a positive velocity, $H_k = H_{k-1} + u_\theta \Delta t + \ldots$, so nothing anchors the level to the observations (Section 4.4). The observation-corrected encoder, PGIB-KF (KF: Kalman filter), therefore adds a level read-out $l_k = f_l(\mathbf{h}_k)$ and a learned gain $\omega_k = \sigma(f_\omega(\mathbf{h}_k)) \in (0, 1)$ to the encoder features $\mathbf{h}_k$,

$$H_k = (1 - \omega_k)\big(H_{k-1} + u_k + s_k\epsilon_k\big) + \omega_k\, l_k. \tag{B.3}$$

Given H_{k-1} this is again a Gaussian transition with drift $u_k^{\text{eff}} = (1 - \omega_k)u_k + \omega_k(l_k - H_{k-1})$ and standard deviation $s_k^{\text{eff}} = (1 - \omega_k)s_k$, so Proposition 6 applies unchanged: the bottleneck now prices the use of the observation beyond what the physics prior allows ($\beta \to \infty$ gives $\omega \to 0$, a pure integrator). The HI is no longer monotone by construction.

The error accumulation of an integrating encoder and its repair by the gain ω are analysed in Proposition 2.

By Proposition 1, a regime-blind read-out has an error floor, and per-regime normalisation removes the offsets but not the gains. The regime-aware variant, *PGIB-KFR*, therefore places the regime in the *observation* model, where the proposition locates it. The process model is shared, whereas the read-out l_k and the gain ω_k use regime-specific linear heads on the encoder features, $l_k = \mathbf{h}_k^\top \mathbf{W}^l_{r_k} + b^l_{r_k}$ and $\omega_k = \sigma(\mathbf{h}_k^\top \mathbf{W}^\omega_{r_k} + b^\omega_{r_k})$, where r_k is the regime, which is also an input of the encoder. The regimes are the clusters of the standardised operating settings. With a single regime PGIB-KFR reduces to PGIB-KF, which we check on the single-regime subsets.

B.4. Calibrated predictive law with age fusion

Because the age is always known (Theorem 1(c)), the HI-based law $p_{\text{HI}}(\tau)$, which is (B.2) with diffusion $c\,v_t$ and drift variance $c\,v_t/w_t$, is pooled log-linearly [70, 71] with an age law. Here v_t is the variance of the HI increments over the trailing window of length w_t for non-learned HIs and the learned σ_B^2 for the bottleneck models, and $c > 0$ is a calibration multiplier. The age law is $p_{\text{age}}(\tau \mid t) = f_L(t+\tau)/S_L(t)$, the log-normal lifetime density [72] fitted on training lifetimes (log-scale floor 0.15) conditioned on survival to age t. The pooled law is

$$p(\tau) \;\propto\; p_{\text{HI}}(\tau)^{w_{\text{h}}}\, p_{\text{age}}(\tau \mid t)^{w_{\text{a}}}, \qquad (c, w_{\text{h}}, w_{\text{a}}) \in \mathcal{G}, \tag{B.4}$$

on a finite grid $\mathcal{G}$ ($c \in [10^{-2}, 10^{2}]$ on a log scale, ten pairs $(w_{\mathrm{h}}, w_{\mathrm{a}})$ including $(1, 0)$ and $(0, 1)$). The parameters minimise the negative log-likelihood of the true RUL, a strictly proper scoring rule [73], on HI paths produced *out of sample*: each HI construction is fitted on all but one training unit and applied to the left-out unit; the test unit is never used. The family contains the plain HI law and the age-only law, so in-sample the pooled law cannot do worse than either. Weights below one temper the corresponding law and widen the predictive interval, which the otherwise under-covering plain law requires. The pooling is a calibration device, not a Bayes-exact fusion, which would also require conditional independence of the HI and the age given the RUL, and division by the marginal law of the RUL. The same predictor is used for every HI, and a variant without fusion ($w_{\mathrm{a}} = 0$) is reported for each HI, so that the effects of the HI and of the fusion can be separated.

C. Data, protocols and evidence tiers

Data. Public run-to-failure collections were used: NASA PCoE cells B0005, B0006, B0007 and B0018 [74]; CALCE CS2 cells [75]; the FEMTO-ST/PRONOSTIA bearings [76]; the NASA IMS bearings [77, 78]; and the C-MAPSS turbofan engines [79]. For batteries, the capacity is used only to define the EOL and is never an input. The EOL is the first cycle at which the rolling-median capacity falls to 75% (NASA) or 80% (CALCE) of the initial capacity; an absolute threshold of 1.4 Ah was not used because cell B0007 never reaches it. The lifetimes are 127, 73, 162 and 100 cycles (NASA) and 542, 499, 572 and 600 cycles (CS2_35 to CS2_38); the data are truncated at the EOL, and CALCE features are averaged over blocks of 4 cycles. Six per-cycle features are extracted from the charge and discharge records: constant-current time, constant-voltage time, height and voltage of the incremental-capacity peak, mean discharge temperature (NASA) or voltage drop at discharge onset (CALCE), and the discharge voltage at a fixed time (600 s for NASA, 1200 s for CALCE).

Further cells were screened by inclusion rules fixed before running any model (scripts and outputs are supplied). A NASA cell had to be at 24°C, have at least 100 discharge cycles, be discharged at a roughly constant 1.4 to 2.1 A, start between 1.5 and 2.1 Ah and reach the 75% EOL; only the four cells above qualify. Among the CALCE cells, CS2_5, 24 and 25 are partial-discharge tests, CS2_9 is discharged at a rate of 2C (twice the nominal capacity per hour), and CS2_3, 6 and 7 are short or follow other protocols, so none was used. CS2_33 and CS2_34 have the same chemistry and run to the EOL (485 and 479 cycles) but are discharged at 0.55 A; they are used only in a separate six-cell experiment, labelled protocol-heterogeneous, and the four-cell set remains the primary one.

For bearings, each run ends at failure, so EOL is the last snapshot. Condition 1 (nominally 1800 rpm, 4000 N; Bearing1_1–1_7) and condition 2 (1650 rpm, 4200 N; Bearing2_1–2_7) are evaluated separately, each with 7 units from the learning and the full test sets; condition 3 has only 3 units and was not used. The lifetimes are 2803, 871, 2375, 1428, 2463, 2448, 2259 snapshots (condition 1) and 911, 797, 1955, 751, 2311, 701, 230 snapshots (condition 2), each snapshot covering 0.1 s of 25.6 kHz acceleration every 10 s. Eight features are computed per snapshot: logarithm of the root mean square (RMS), kurtosis, crest factor and spectral entropy of the horizontal and of the vertical channel. Features are averaged over blocks of 20 (condition 1) or 8 (condition 2) snapshots, and the RUL is reported in snapshots.

Evaluation is leave-one-unit-out: each unit in turn is held out, the remaining units are used for training (three for NASA and four-cell CALCE, five for six-cell CALCE, six for

each bearing condition), and metrics are computed at every step in the window 40–90% of life and pooled over the held-out units.

Further collections. FEMTO condition 3 (1500 rpm, 5000 N; Bearing3_1–3_3, with 515, 1637 and 434 snapshots, block length 10) and the NASA IMS bearings were kept aside and used only as confirmation sets. For IMS the four bearings that failed (Test 1 bearings 3 and 4, Test 2 bearing 1, Test 3 bearing 3; 2156, 2156, 984 and 6324 snapshots of 1 s at 20 kHz every 10 min) are used with one accelerometer channel each and four features per snapshot (log RMS, kurtosis, crest factor, spectral entropy), averaged over blocks of 20 snapshots. For C-MAPSS we use the single-condition subsets FD001 and FD003 (100 training engines run to failure and 100 truncated test engines with the official RUL labels each); the 14 informative sensors are standardised with training statistics. The 100 training engines are split by identifier into 80 engines to fit the HI constructions and 20 (identifier divisible by five) to select the bottleneck configuration and to calibrate the predictor; the test engines are predicted once at their last observed cycle, and 95% bootstrap intervals are computed over engines. The multi-regime subsets FD002 (260 training and 259 test engines) and FD004 (249 and 248) have six operating regimes; for them the only change is that sensors are standardised within regime, the six regimes being the clusters of the standardised operating settings found on the fitting engines (the clusters recover the six discrete settings exactly), the same preprocessing being applied to every method; the split is by identifier as above, with 208 and 200 fitting engines.

Methods. To isolate the effect of HI construction, every HI is mapped to the same scale and passed to the same calibrated predictor of Appendix B.4. The sensor features are standardised, and each HI is mapped by an affine map $H \mapsto (H - H_{\mathrm{s}})/(H_{\mathrm{e}} - H_{\mathrm{s}})$, where H_{s} and H_{e} are the averages, over the *training* units only, of the mean of the first five and of the last five HI values of a unit (the end of life of a training unit is known); the standardisation, the PCA direction, the combination weights of the Geo-opt HI, the encoders and the affine map are all fitted on the training units and applied unchanged to the test unit, whose own end-of-life value is never used. The PGIB scale is set by its end-of-life penalty on the training units. The predictor of Appendix B.4 then uses: the mixed inverse-Gaussian law (B.2) with the drift taken from the slope of the smoothed HI over a trailing window of $\max(10, 0.25t)$ steps, pooled with the age law. The parameters $(c, w_{\mathrm{h}}, w_{\mathrm{a}})$ are chosen on HI paths that are out of sample for the HI construction (leave-one-training-unit-out for the small sets, the 20 calibration engines for C-MAPSS). Implementation details of the compared HIs: the Geo-opt HI is found by differential evolution; the β-VAE uses a causal gated recurrent unit (GRU) encoder with a one-dimensional latent variable and a Gaussian prior; VIB is the PGIB with a zero-drift prior, that is, without physics; the PGIB has the state-dependent drift $\eta(H; \vartheta) = \vartheta_1 \exp(\vartheta_2 H)$ (Appendix B.3 for PGIB-KF). For the bottleneck models, the compression weight $\beta \in \{0.03, 0.3, 3, 30\}$ and the encoder (GRU or causal feed-forward window network) are selected by inner leave-one-training-unit-out validation of the predictive negative log-likelihood, with at most three inner folds when more than five training units are available; for PGIB-KF only $\beta \in \{0.03, 0.3, 3\}$ is searched. The held-out unit is never used. Learned models use three seeds, and the other HIs are deterministic. The sensor-free reference is the age law p_{age} of Appendix B.4. The metrics are the RMSE and the mean absolute error (MAE) of the median prediction, the prediction interval coverage probability (PICP) and normalised average width (PINAW) of the 95% interval, and the continuous ranked probability score (CRPS) [73]. For C-

Table C.6: Algorithm 1: training of the PGIB and Bayesian FHT prognostics (Appendix B).

Input: training sequences $\{\mathbf{X}_{1:T_n}^{(n)}, T^{(n)}\}$, threshold $D_{\text{fail}} = 1$, compression weight β, step Δt. **Output:** q_{θ^*}, $(\vartheta^*, \sigma_B^*)$, predictive law (B.2).
1. Causal encoder (temporal convolutional network or GRU): $u_\theta, s_\theta \leftarrow \text{softplus}(\cdot)$ from $\mathbf{X}_{1:k}$.
2. Sample the path by reparametrisation: $H_k = H_{k-1} + u_\theta \Delta t + s_\theta \sqrt{\Delta t}\, \epsilon_k$, $\epsilon_k \sim \mathcal{N}(0,1)$, $H_0 = 0$.
3. Conjugate drift update from $\Delta H_{1:t}$ to obtain $(\hat{\mu}_t, \sigma_{\mu,t}^2)$.
4. $\mathcal{L}_{\text{FHT}} = -\ln p(T_t \mid H_{1:t})$ from (B.2); $\mathcal{L}_{\text{KL}}$ from Proposition 6.
5. Update $(\theta, \vartheta, \sigma_B)$ with the AdamW optimiser on $\mathcal{L}_{\text{FHT}} + \beta \mathcal{L}_{\text{KL}}$ (plus a penalty on $H_{T_{\text{EOL}}} - D_{\text{fail}}$ at the EOL).

MAPSS the asymmetric score is $\sum_i (e^{-d_i/13} - 1)$ for $d_i < 0$ and $\sum_i (e^{d_i/10} - 1)$ for $d_i \geq 0$, summed over engines, with d_i the predicted minus the true RUL [79].

Protocol history and evidence tiers. The study proceeded in five protocols, which determine how the numbers can be read. P1 (Appendix E) was the first benchmark; it showed that the integrating encoder under-covers and that the age law is competitive. P2 introduced the resulting improvements (out-of-sample calibration, age fusion, a prediction grid independent of the test unit) and was frozen, with the file hash recorded, before it was run on the two confirmation sets (FEMTO condition 3 and IMS) and before the development sets were re-run. On C-MAPSS FD001 the P2 methods showed that the integrating encoder is considerably worse than simple HIs even with 80 training engines. The diagnosis of Proposition 2 led to P3, the observation-corrected PGIB-KF, which was frozen and run once on FD003, a subset not used before, and exploratorily on the small sets. The unchanged P3 model was then frozen with regime-wise preprocessing and run once on FD002 and FD004. (A first launch used only the first 100 training engines because the identifier list was hard-coded in the script; it was stopped before any result was produced, and the script and the recorded hash were corrected.) After these results, P4 added the regime-aware encoder and a regime-specific ridge baseline and was run once on each subset, the choice between PGIB-KF and PGIB-KFR and of β being made on the calibration engines only. Because FD002 and FD004 had already been seen, they are development sets for P4, and no independent multi-regime data set was available (the successor N-CMAPSS has only six to ten units per subset). Finally, because the differences between learned encoders and simple HIs were of the size of the seed-to-seed variability, P5 retrained the learned encoders on the four subsets with ten fresh seeds, keeping architecture, β and calibration fixed; it supersedes the three-seed results where they disagree. The evidence tiers are therefore: (i) *clean confirmation*: C-MAPSS FD003, FD002 and FD004 for P3 (not for P4), and FEMTO condition 3 and IMS for P2 (not for P3, whose idea followed their P2 results); (ii) *development*: NASA, CALCE (four and six cells), FEMTO conditions 1 and 2 and C-MAPSS FD001, which informed the design and whose results are therefore not unbiased estimates of out-of-sample performance.

D. Detailed benchmark results

This appendix gives the tables behind Section 6: protocol P2 on the development and confirmation sets (Tables D.7 and D.8), the C-MAPSS subsets (Tables D.9, D.10,

D.11 and D.12), the effect of the age fusion (Table D.13), an ablation of the integrating PGIB (Table D.14) and the certified information (Tables D.15 and D.16). Protocols and evidence tiers are described in Appendix C.

E. Initial benchmark protocol (P1)

The first benchmark used a simpler protocol (P1). Every HI was calibrated on in-sample training paths, the plain first-hitting-time law (B.2) was used without age fusion, and the predictive grid extended to three times the held-out unit's own life, which leaks the length of the test unit into the support of the predictive law (removed in P2). The pilot on NASA B0005, the results below and the observation that the age law is competitive motivated P2. The P1 results are reported for completeness and are not used for the conclusions of the main text.

Pilot run. A pilot on the NASA B0005 fold, with a GRU-only encoder and $\beta \in \{0.003, 0.03, 0.3\}$, gave a PGIB RMSE of 56.3 cycles and zero interval coverage, with the selected β at the edge of the grid. The selection grid was therefore extended and the feed-forward encoder added as a candidate. Both changes are chosen per fold by inner validation, but the decision to introduce them was made after seeing a held-out result for one fold. No baseline was modified, and no change was made after the full runs reported below; the bearing experiments were run once with the final protocol.

Results under P1. The outcome depends on the data set (Tables E.17 and E.18, Figure E.10), and the age law is needed to read it. On NASA the age-law RMSE is 39.8 cycles; PCA-HI (17.8), the Geo-opt HI (21.9), the β-VAE (30.8) and VIB (33.3) beat it, whereas the PGIB (47.6) does not, and its intervals cover only 42% of the true RULs. On CALCE the age-law RMSE is 49.3; only the PGIB (33.0) beats it clearly, while PCA-HI, the Geo-opt HI and the β-VAE (117 to 164) are considerably worse than the age law. With the two 0.5C-discharge cells added, the PGIB keeps the lowest RMSE (50.4) but its margin over the age law (54.7) is only 8%. On bearings the age-law RMSE (602 and 664 snapshots for conditions 1 and 2) is comparable to the standard deviation of the pooled RUL, and most HIs do no better: the PGIB has the lowest RMSE in condition 1 (524.5, 13% below the age law) but gives almost no benefit in condition 2 (621.8 against 664). Two long-lived units dominate the pooled error in condition 2.

The ablations (in the supplied result files) support the bottleneck on all five sets (for example RMSE 54.4 $\to$ 47.6 on NASA and 42.3 $\to$ 33.0 on CALCE) and the epistemic drift variance for coverage (for example 14 $\to$ 42% on NASA). The physics prior helps on CALCE and FEMTO condition 1 but not on NASA or FEMTO condition 2. With only three to six training units per fold, the validation signal that selects the encoder is itself noisy; on NASA it chose the strongest compression ($\beta = 30$), at the edge of the grid, in all eight selections.

Geometric scores versus error. Across the five HIs of each data set, the rank correlation between the geometric score Mon + Tre + Pro and the RMSE is +0.90 on NASA (a higher geometric score goes with a larger error), −1.00 on CALCE, −0.90 on six-cell CALCE, −0.60 on FEMTO condition 1 and +0.60 on condition 2. The geometric criteria therefore did not rank the HIs consistently by prognostic error. This is an association over five methods per data set and not evidence for the mechanism of Theorem 3: the VIB and

Table D.7: Protocol P2, development sets. RMSE, MAE and CRPS in cycles (batteries) or 10 s snapshots (bearings); mean ± standard deviation over seeds; bold: lowest RMSE.

NASA PCoE (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	34.4	28.7	84	2.68	21.0
PCA-HI	24.9	19.3	95	2.46	13.8
Geo-opt HI	**24.3**	20.1	93	3.22	17.6
β-VAE-HI	31.7±2.2	25.3±1.8	98±0	4.24±0.17	20.0±0.9
VIB-HI (no physics)	33.9±1.2	28.3±1.2	90±1	5.40±0.03	27.1±0.7
PGIB-HI (ens.)	34.8	29.0	93	6.51	27.3
PGIB-KF-HI (ens.)	44.3	39.4	84	4.55	29.1

CALCE CS2 (4 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	46.5	40.0	100	1.13	28.6
PCA-HI	51.3	44.0	100	1.06	30.2
Geo-opt HI	43.3	35.7	98	0.98	26.5
β-VAE-HI	49.9±0.9	41.6±1.0	100	1.09±0.00	29.5±0.4
VIB-HI (no physics)	48.3±1.4	40.6±0.9	83±4	0.85±0.00	30.8±0.8
PGIB-HI (ens.)	**38.4**	34.2	99	0.74	23.3
PGIB-KF-HI (ens.)	50.3	36.6	77	0.83	33.0

CALCE CS2 (6 cells)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	50.6	45.5	100	1.13	30.1
PCA-HI	56.8	50.0	100	1.09	32.8
Geo-opt HI	**46.3**	43.2	100	1.02	27.6
β-VAE-HI	73.5±4.9	60.9±3.9	93±3	1.04±0.02	42.4±3.2
VIB-HI (no physics)	54.7±1.7	48.7±1.5	92±1	0.91±0.01	32.4±1.2
PGIB-HI (ens.)	48.9	44.8	90	0.87	30.8
PGIB-KF-HI (ens.)	71.9	63.6	84	0.92	45.2

FEMTO cond. 1 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**495.8**	371.4	94	3.45	303.7
PCA-HI	786.9	621.6	72	4.72	489.2
Geo-opt HI	533.2	416.8	96	3.19	311.5
β-VAE-HI	737.1±191.7	525.4±102.0	85±6	4.84±0.04	421.4±87.9
VIB-HI (no physics)	584.1±29.4	467.0±26.4	82±5	3.48±0.13	357.9±29.0
PGIB-HI (ens.)	534.7	401.8	84	4.67	340.9

FEMTO cond. 2 (7 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	**430.6**	373.5	97	10.38	278.6
PCA-HI	558.3	450.5	67	8.52	391.1
Geo-opt HI	504.9	414.7	85	11.22	396.1
β-VAE-HI	531.2±11.3	434.6±20.1	88±2	9.78±0.02	368.4±2.0
VIB-HI (no physics)	448.2±0.4	337.7±1.3	97	11.39±0.14	318.5±2.6
PGIB-HI (ens.)	449.5	333.6	97	13.97	341.2

Table D.8: Protocol P2, confirmation sets (snapshots). PGIB-KF is exploratory here.

FEMTO cond. 3 (3 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	563.6	529.7	37	10.60	460.2
PCA-HI	507.5	418.4	32	6.50	398.8
Geo-opt HI	567.1	525.1	30	12.82	453.4
β-VAE-HI	**462.0±20.8**	389.5±23.6	37	8.86±0.72	363.4±20.9
VIB-HI (no physics)	505.0±0.1	438.3±0.3	37	12.55±0.00	442.5±0.2
PGIB-HI (ens.)	510.3	450.0	37	12.56	453.5
PGIB-KF-HI (ens.)	555.0	479.3	33	9.10	438.6

NASA IMS (4 bearings)					
Method	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only (no sensors)	1692.8	1503.1	83	10.78	1106.3
PCA-HI	**745.2**	605.7	96	8.05	498.0
Geo-opt HI	942.7	702.8	99	16.31	771.6
β-VAE-HI	2769.6±282.2	1324.8±242.7	84±6	10.31±0.71	1184.6±197.8
VIB-HI (no physics)	1884.6±10.5	1434.4±12.4	91	16.41±0.01	1122.5±8.8
PGIB-HI (ens.)	1623.8	1288.1	98	16.49	1017.5
PGIB-KF-HI (ens.)	1852.3	1519.8	43	12.29	1545.6

Table D.9: C-MAPSS FD003 (clean confirmation set of P3; PGIB-KFR and regime-specific ridge rows are development results). RMSE: single models, mean over seeds; seed-avg.: RMSE of the seed-averaged prediction with 95% bootstrap interval over engines.

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	47.6	42.3–52.6	39.6	114536	99	1.91	28.6
PCA-HI	33.8	28.4–38.3	25.5	10747	97	1.47	19.7
Geo-opt HI	21.9	17.8–26.0	15.9	1864	100	1.19	13.1
Ridge-HI (supervised)	26.8	22.9–30.6	21.0	6147	93	1.20	16.2
Ridge-HI, regime-specific	26.8	22.8–30.6	21.0	6147	93	1.20	16.2
β-VAE-HI	40.6±1.1	34.9–45.6	32.0±0.7	29236	99	1.69	23.8±0.3
VIB-HI (no physics prior)	37.9±0.8	31.7–42.2	29.3±0.8	18680	96	2.16	24.2±0.6
PGIB-HI (GRU/window encoder)	39.9±1.0	33.9–44.4	31.2±0.4	21945	95	1.98	24.3±0.3
PGIB-HI, 3-seed ensemble	39.5	34.1–45.0	31.1	17643	95	1.98	24.1
PGIB-KF-HI	24.2±2.5	16.4–23.1	18.6±1.5	2296	94	1.19	14.8±0.4
PGIB-KF-HI, 3-seed ensemble	19.4	16.0–22.8	14.6	790	97	1.20	12.2
PGIB-KFR-HI (regime-aware)	27.4±0.1	21.8–29.3	21.2±0.3	5417	95	1.24	15.6±0.4
PGIB-KFR-HI, 3-seed ensemble	25.4	21.6–29.1	20.3	4170	97	1.24	14.6

Table D.10: C-MAPSS FD001 (development set; same layout as Table D.9).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	36.6	32.4–40.8	30.5	8023	99	1.18	21.1
PCA-HI	21.8	19.1–24.4	17.9	908	96	0.83	12.6
Geo-opt HI	22.0	19.5–24.3	18.0	953	96	0.83	12.7
Ridge-HI (supervised)	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
Ridge-HI, regime-specific	26.0	22.0–29.8	19.7	1606	97	0.85	14.1
β-VAE-HI	22.2±0.4	19.0–24.5	18.0±0.5	875	95	0.82	12.7±0.3
VIB-HI (no physics prior)	34.2±0.3	28.8–38.3	27.1±0.7	7386	91	0.96	19.3±0.4
PGIB-HI (GRU/window encoder)	35.6±1.0	29.8–40.9	26.9±0.6	11524	94	0.90	19.6±0.4
PGIB-HI, 3-seed ensemble	35.4	29.9–40.9	26.6	10874	94	0.90	19.4
PGIB-KF-HI	27.0±2.2	18.8–27.1	20.2±1.2	2868	82	0.56	14.9±1.1
PGIB-KF-HI, 3-seed ensemble	22.9	18.6–26.9	16.6	1309	93	0.56	12.1
PGIB-KFR-HI (regime-aware)	26.9±3.6	21.8–31.0	20.2±2.6	2937	94	0.64	14.4±2.1
PGIB-KFR-HI, 3-seed ensemble	26.5	22.0–30.9	19.5	2094	96	0.64	13.8

Table D.11: C-MAPSS FD002 (multi-regime; confirmation set for P3, development set for P4; same layout as Table D.9).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	38.3	35.2–41.2	31.4	53131	97	0.82	21.6
PCA-HI	28.7	26.0–31.4	21.9	10331	95	0.66	15.7
Geo-opt HI	30.5	27.0–33.9	21.8	16164	98	0.59	15.2
Ridge-HI (supervised)	29.4	26.1–32.6	21.7	11935	94	0.58	15.1
Ridge-HI, regime-specific	28.4	25.3–31.5	20.9	11149	94	0.58	14.7
β-VAE-HI	30.6±0.8	27.3–33.1	24.0±0.7	11996	92	0.65	17.0±0.4
VIB-HI (no physics prior)	36.4±0.4	33.3–39.4	29.8±0.2	26118	82	0.68	21.1±0.1
PGIB-HI (GRU/window encoder)	36.8±0.2	33.6–39.4	30.1±0.5	28553	83	0.68	21.4±0.5
PGIB-HI, 3-seed ensemble	36.6	33.7–39.4	29.9	26625	82	0.68	21.3
PGIB-KF-HI	31.3±1.5	27.8–33.1	25.4±1.5	14139	81	0.79	18.6±1.1
PGIB-KF-HI, 3-seed ensemble	30.4	27.8–32.9	24.5	10983	81	0.79	18.0
PGIB-KFR-HI (regime-aware)	30.7±2.0	27.2–32.2	24.8±2.6	14802	86	0.91	18.5±1.9
PGIB-KFR-HI, 3-seed ensemble	29.6	27.0–32.1	23.7	13089	87	0.91	17.8

Table D.12: C-MAPSS FD004 (multi-regime; confirmation set for P3, development set for P4; same layout as Table D.9).

Method	RMSE	RMSE seed-avg. [95% CI]	MAE	Score	PICP [%]	PINAW	CRPS
Age-only (no sensors)	54.7	50.9–58.5	46.0	611896	95	1.31	31.8
PCA-HI	44.3	40.7–48.0	35.4	104864	92	1.03	24.9
Geo-opt HI	29.7	26.6–32.7	21.9	9468	100	0.86	16.1
Ridge-HI (supervised)	30.9	28.1–33.4	23.8	11632	94	1.14	18.5
Ridge-HI, regime-specific	31.1	28.2–33.9	24.0	10794	92	1.14	18.6
β-VAE-HI	48.1±3.1	42.7–50.2	39.0±3.0	294449	93	1.17	27.7±1.9
VIB-HI (no physics prior)	51.8±1.0	47.3–56.2	41.9±0.6	697542	90	1.13	29.5±0.4
PGIB-HI (GRU/window encoder)	49.9±2.4	44.2–53.9	39.5±1.5	1143425	90	1.42	29.3±0.7
PGIB-HI, 3-seed ensemble	49.5	44.8–54.3	38.9	1085286	91	1.42	28.9
PGIB-KF-HI	42.1±2.8	37.2–43.6	33.9±1.9	241870	78	1.08	25.6±1.3
PGIB-KF-HI, 3-seed ensemble	40.6	37.3–44.0	32.9	124580	79	1.07	24.7
PGIB-KFR-HI (regime-aware)	37.1±2.5	31.4–37.2	29.4±1.5	187056	85	1.12	22.3±0.8
PGIB-KFR-HI, 3-seed ensemble	34.8	31.8–37.8	27.8	37063	85	1.12	21.0

Table D.13: Effect of the age fusion: RMSE with fusion / without fusion (after recalibration with $w_a = 0$).

HI	NASA	CALCE-4	CALCE-6	FEMTO-1	FEMTO-2	FEMTO-3	IMS
PCA-HI	25 / 18	51 / 186	57 / 184	787 / 1470	558 / 564	507 / 503	745 / 745
Geo-opt HI	24 / 32	43 / 131	46 / 126	533 / 1083	505 / 489	567 / 437	943 / 943
β-VAE-HI	32 / 26	50 / 110	73 / 270	737 / 1421	531 / 457	462 / 396	2770 / 2770
VIB-HI (no physics)	34 / 30	48 / 59	55 / 106	584 / 681	448 / 469	505 / 507	1885 / 1885
PGIB-HI (ens.)	35 / 30	38 / 35	49 / 85	535 / 754	449 / 445	510 / 509	1624 / 1624
PGIB-KF-HI (ens.)	44 / 44	50 / 59	72 / 88	–	–	555 / 556	1852 / 1852

Table D.14: Ablation of the integrating PGIB on C-MAPSS FD003 (age-fused predictor; mean ± standard deviation over three seeds). PICP, PINAW: coverage and normalised width of the 95% interval; CRPS: continuous ranked probability score.

Variant	RMSE	MAE	PICP [%]	PINAW	CRPS
Full PGIB	39.9±1.0	31.2±0.4	95±1	1.98±0.03	24.3±0.3
Without bottleneck ($\beta = 0$)	39.1±1.9	30.7±1.0	96±0	1.99±0.04	23.9±0.7
Without physics prior (VIB)	37.9±0.8	29.3±0.8	96±0	2.16±0.05	24.2±0.6
Without epistemic drift variance	45.9±1.4	37.3±1.4	92±1	1.32±0.09	26.1±0.8

Table D.15: Protocol P2, information certified by the achieved error (nats); < 0: vacuous. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB	PGIB-KF
NASA	4.36	< 0	< 0	< 0	< 0	< 0	< 0	< 0
CALCE-4	5.67	0.41	0.31	0.48	0.34	0.37	0.60	0.33
CALCE-6	5.66	0.31	0.20	0.40	< 0	0.24	0.35	< 0
FEMTO-1	7.24	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-2	7.04	< 0	< 0	< 0	< 0	< 0	< 0	–
FEMTO-3	6.70	< 0	< 0	< 0	< 0	< 0	< 0	< 0
IMS	8.05	< 0	0.02	< 0	< 0	< 0	< 0	< 0

Table D.16: Age-conditional certificate (nats) under a fitted log-normal lifetime model; $\bar{h}(T \mid t)$: mean conditional entropy. A positive value for the age-only predictor shows that the model overestimates $h(T \mid t)$.

Data set	$\bar{h}(T \mid t)$	Age-only	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.67	-0.28	0.04	0.07	-0.14	-0.23	-0.29
CALCE-4	5.74	0.49	0.39	0.56	0.42	0.46	0.68
CALCE-6	5.69	0.35	0.23	0.44	-0.05	0.31	0.38
FEMTO-1	7.81	0.19	-0.28	0.11	-0.15	-0.03	0.11
FEMTO-2	7.77	0.29	0.03	0.13	0.08	0.25	0.24
FEMTO-3	7.30	-0.45	-0.35	-0.46	-0.19	-0.34	-0.35
IMS	8.65	-0.20	0.62	0.39	-0.58	-0.31	-0.16

Table E.17: Protocol P1, battery benchmarks, leave-one-unit-out (mean ± standard deviation over seeds). RMSE, MAE and CRPS in cycles.

NASA PCoE (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	39.8	34.7	–	–	–
PCA-HI	0.27	0.85	0.77	17.8	14.0	97	2.52	11.1
Geo-opt HI	0.21	0.93	0.95	21.9	17.0	90	2.43	12.8
β-VAE-HI	0.30±0.01	0.71±0.00	0.95±0.01	30.8±0.8	26.0±0.5	95±1	2.06±0.08	17.6±0.5
VIB-HI (no physics prior)	1.00	1.00±0.00	0.67±0.03	33.3±2.8	27.7±2.1	88±4	3.47±0.10	20.5±2.3
PGIB-HI	1.00	1.00±0.00	0.67±0.01	47.6±2.1	42.3±1.9	42±3	2.39±0.15	34.3±1.5
CALCE CS2 (4 cells, 1C/1C) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	49.3	43.8	–	–	–
PCA-HI	0.05	0.72	0.83	130.5	110.4	92	2.74	82.3
Geo-opt HI	0.39	0.95	0.98	117.2	86.8	90	3.20	69.2
β-VAE-HI	0.02±0.01	0.72±0.01	0.80±0.01	163.7±0.7	148.0±0.9	83±1	1.87±0.02	111.1±0.7
VIB-HI (no physics prior)	1.00	1.00±0.00	0.92±0.01	53.2±0.9	47.2±1.4	71±4	1.24±0.13	38.4±0.7
PGIB-HI	1.00	1.00±0.00	0.95±0.01	33.0±4.9	29.9±3.9	55±8	0.34±0.02	22.6±3.1
CALCE CS2 (6 cells, incl. 0.5C-discharge CS2_33/34) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	54.7	50.0	–	–	–
PCA-HI	0.13	0.28	0.21	191.8	172.0	50	0.99	149.6
Geo-opt HI	0.39	0.95	0.98	106.9	82.7	95	3.11	63.7
β-VAE-HI	0.05±0.02	0.52±0.10	0.33±0.04	201.6±5.3	179.9±5.0	48±0	1.46±0.18	156.1±3.1
VIB-HI (no physics prior)	1.00	1.00±0.00	0.87±0.02	69.7±6.0	54.4±8.8	90±5	2.24±0.23	44.9±4.8
PGIB-HI	1.00	1.00±0.00	0.91±0.01	50.4±4.4	44.7±5.1	69±5	0.58±0.02	33.4±4.2

Table E.18: Protocol P1, FEMTO bearing benchmarks. RMSE, MAE and CRPS in 10 s snapshots.

FEMTO bearings, condition 1 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	602.5	515.4	–	–	–
PCA-HI	0.06	0.56	0.70	784.5	663.2	93	2.80	485.7
Geo-opt HI	0.04	0.29	0.39	770.2	623.3	91	3.34	459.1
β-VAE-HI	0.07±0.02	0.58±0.01	0.73±0.01	779.8±51.1	664.4±35.6	91±3	2.69±0.13	492.4±20.6
VIB-HI (no physics prior)	1.00	1.00±0.00	0.71±0.02	570.1±56.0	471.0±50.7	78±5	3.04±0.21	365.6±44.1
PGIB-HI	1.00	1.00±0.00	0.71±0.02	524.5±45.3	423.6±37.3	75±5	2.65±0.10	336.5±31.8
FEMTO bearings, condition 2 (7 units) (leave-one-unit-out)								
Method	Mon	Tre	Pro	RMSE	MAE	PICP [%]	PINAW	CRPS
Age-only reference (no sensors)	–	–	–	664.4	584.6	–	–	–
PCA-HI	0.04	0.43	0.17	613.9	487.6	54	2.25	433.9
Geo-opt HI	0.06	0.37	0.46	559.2	457.1	73	2.33	356.5
β-VAE-HI	0.06±0.01	0.46±0.05	0.42±0.16	539.8±31.7	439.6±24.6	70±2	2.01±0.06	348.6±26.1
VIB-HI (no physics prior)	1.00	0.99±0.00	0.39±0.02	616.8±1.5	493.8±1.5	41±1	2.62±0.23	451.1±3.2
PGIB-HI	1.00	0.99±0.00	0.46±0.02	621.8±4.3	498.2±8.1	36±2	2.25±0.01	464.3±7.9

Table E.19: Protocol P1, information certified by the achieved error (Theorem 1(b)), in nats; < 0: vacuous. $h(T)$: entropy of the pooled held-out RUL.

Dataset	$h(T)$	PCA	Geo-opt.	β-VAE	VIB	PGIB
NASA	4.36	0.06	< 0	< 0	< 0	< 0
CALCE (4)	5.67	< 0	< 0	< 0	0.27	0.76
CALCE (6)	5.66	< 0	< 0	< 0	< 0	0.32
FEMTO c. 1	7.24	< 0	< 0	< 0	< 0	< 0
FEMTO c. 2	7.04	< 0	< 0	< 0	< 0	< 0

PGIB HIs have Mon $\approx$ Tre ≈ 1 because their velocity is constrained to be positive, not because of a squeeze.

Information on real data. A direct estimate of $I(T; H)$ with the Kraskov–Stögbauer–Grassberger estimator [48] on rank-transformed window summaries gave, for several HIs, values above the estimate for the 12-dimensional raw sensor summary, which is impossible for a true mutual information because the summary contains the HI. This reflects the bias of the estimator in 12 dimensions at a few hundred samples, a known limitation [49, 50]. A simulation with known mutual information (twelve Gaussian sensor features, a scalar target and a scalar HI with $\eta_{\text{info}} = 0.7$, Table E.20) shows that the estimator underestimates $I(T; \mathbf{X})$ strongly, so that $\hat{\eta}_{\text{info}}$ exceeds 1 in every replication even with 3000 samples. We therefore do not report η_{info} on real data. The certificate of Theorem 1(b) avoids high-dimensional estimation (Table E.19); it is positive only for the PGIB on CALCE (0.76 nats) and six-cell CALCE (0.32), VIB on CALCE (0.27) and PCA-HI on NASA (0.06), and vacuous otherwise. This is a more modest reading of the benchmark than the RMSE tables alone.

Table E.20: Estimation of η_{info} with the Kraskov–Stögbauer–Grassberger estimator ($k = 5$, 100 replications) in a Gaussian model with known mutual information; n is the sample size.

n	$\hat{I}(T; \mathbf{X})$	$\hat{I}(T; H)$	$\hat{\eta}_{\text{info}}$ (mean $\pm$ standard deviation)	fraction $\hat{\eta}_{\text{info}} > 1$
100	0.23	0.59	2.72 ± 0.74	1.00
300	0.33	0.68	2.10 ± 0.25	1.00
1000	0.43	0.71	1.65 ± 0.09	1.00
3000	0.51	0.71	1.40 ± 0.04	1.00

true values: $I(T; \mathbf{X}) = 1.00$, $I(T; H) = 0.70$, $\eta_{\text{info}} = 0.70$ (nats)

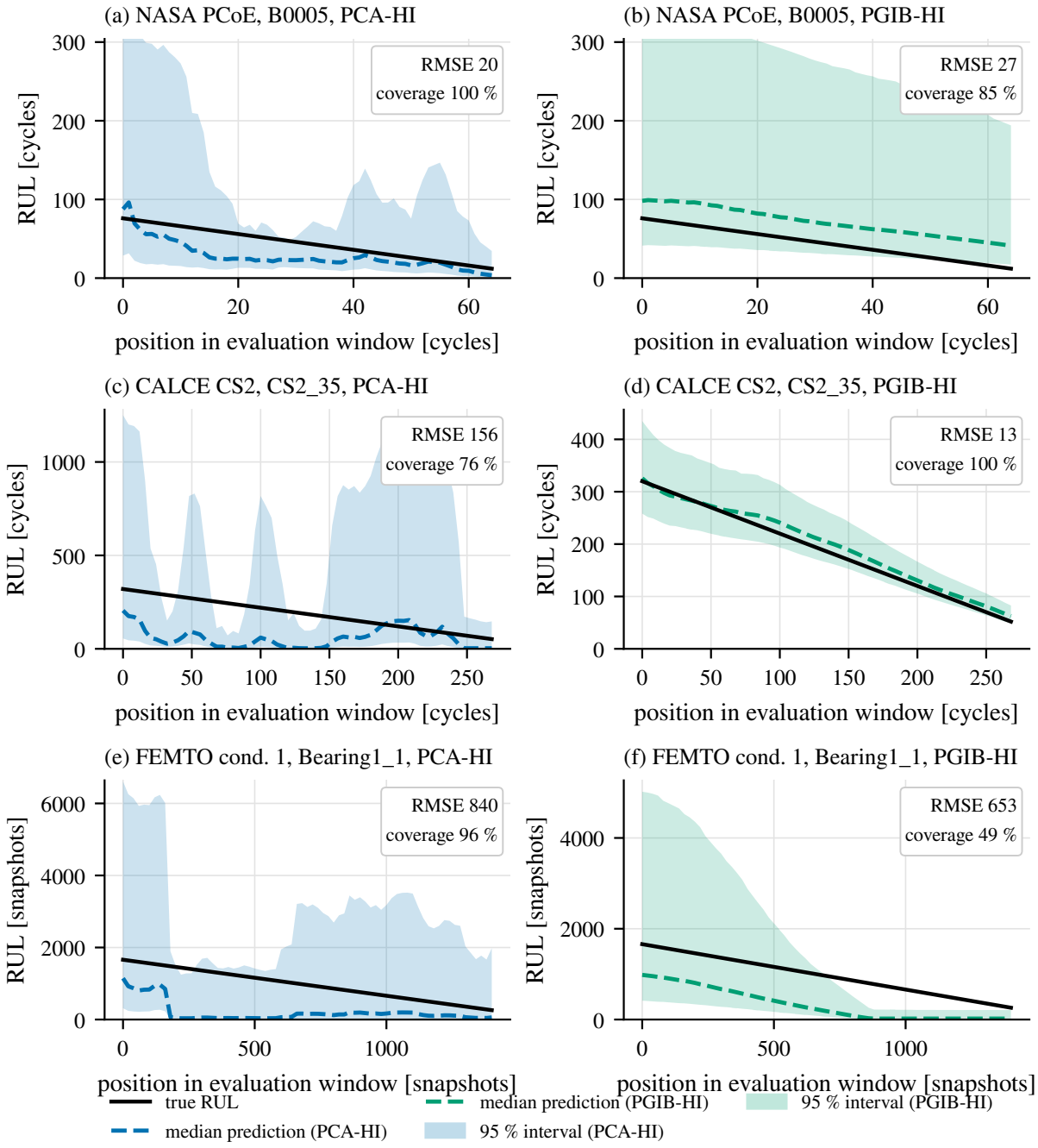


Figure E.10: Example RUL predictions with 95% intervals on one held-out unit per data set (NASA B0005, CALCE CS2_35, FEMTO Bearing1_1), for PCA-HI (left) and PGIB (right); each panel gives the RMSE and the coverage on that unit.

F. Details of the real-data tests

Turbofan engines: set-up. Each subset uses its training engines (run to failure); the 14 informative sensors are standardised with the training engines of the fold, within regime for FD002 and FD004 (six regimes from k-means clustering of the operating settings). The ridge HI is a ridge regression (penalty 10) of the life fraction on the standardised sensors, smoothed by a trailing mean of 5 cycles and mapped affinely to zero at the start and one at the end of life with the training engines of the fold; PCA-HI is the first principal

component. The trajectory summary of an HI consists of its trailing mean over 10 cycles (level), its least-squares slope over the last 30 cycles (velocity), its values 20 and 50 cycles earlier, and the age; the raw-sensor predictor uses level and slope of every sensor and the age. Predictors are histogram-based gradient-boosted regression trees (200 iterations, learning rate 0.08, 31 leaves, at least 40 samples per leaf, L_2 penalty 1) trained on all pairs of the training engines of a fold. The lifetime law is a Gaussian kernel density of $\ln L$ (Scott bandwidth) of the training engines of the fold; the conditional density of the RUL at age t is $f_L(t+\tau)/S_L(t)$ and $h(T_t \mid t)$ is evaluated numerically on a grid of 0.5 cycles. Bootstrap intervals resample engines (300 replicates).

Lithium-ion cells. Each cycle of the processed records yields the discharge capacity, the charge duration (samples with current above 0.01 A) and the charge voltage when the charged capacity first reaches 27% and 73% of the nominal capacity; the HIs use the last three quantities. The charge duration is divided by its mean over cycles 2 to 11 of the same cell and the voltages are reduced by their means over these cycles, and the result is standardised with the training cells of the fold. Time scales are set by the lifetimes. For HUST the minimum age is 100 cycles, ages are sampled every 5 cycles, the windows are 20 cycles (level) and 100 cycles (velocity), the lags are 100 and 250 cycles and the grid of the conditional entropy has a step of 2 cycles. For MATR the corresponding values are 50, 3, 10, 50, 50 and 125 cycles and a step of 1 cycle. The trees have at most 15 leaves, and all other settings are those of the engines. The capacity reference uses level and velocity of the relative discharge capacity and the age. The end of life is the first cycle at which the running median of the discharge capacity over 5 cycles falls below 80% of the mean over the first 5 cycles (HUST), or the life label of the source, at which the recordings end (MATR; recordings that end more than 10 cycles before the label would be excluded, and none was). For Tongji the minimum age is 30 cycles, ages are sampled every 2 cycles, the windows are 10 and 40 cycles, the lags 40 and 100 cycles and the step 1 cycle, and the end of life is the first cycle at which the running median of the discharge capacity falls below 80% of the nominal capacity; cells that do not reach it, or reach it within 100 cycles, are excluded. The capacity levels of the charge-voltage and discharge-voltage features are 3/11 and 8/11 of the nominal capacity. The processed records are unpickled with a restricted reader that admits only array types. The protocol read-out of Table F.24 uses the three charging features standardised over the population (blind) or relative to the own cycles 2 to 11 (normalised), with the ridge read-out of the engines.

Squeeze. The template is the mean ridge HI at each cycle over the training engines of the fold, and the noise standard deviation is $\sigma_2 = \Delta m/(\sqrt{2}\cdot 1.645)$ with Δm the mean template increment over the first half of the median lifetime. Mon_∞ is predicted as the mean of $2\Phi_{\mathcal{N}}(\Delta m_k/(\sqrt{2}\sigma_2)) - 1$ over the template increments Δm_k up to the median lifetime, where $\Phi_{\mathcal{N}}$ is the standard normal distribution function. Because the template slope varies along life, the predicted limit is lower than for a constant slope.

Stages. Level and velocity are the summaries of the out-of-fold ridge HI, standardised over all pairs; the target is the RUL. Stages are bins of the life fraction (0.2 to 1, edges $0.2, 0.35, 0.5, 0.65, 0.8$; oracle stages) or equal-count bins of the age (observable stages). Within a stage, the free direction is the linear least-squares fit on both features, level-only and velocity-only use one feature, and a static direction (grid of 5°) is a fixed combination of the two features with a stage-specific affine map; the static direction minimises the sum

of the out-of-fold MSEs over stages. The tables give the worst-stage RMSE ratio to the free direction.

Official test engines. The certificate uses the observed cycles of each test engine as its age, the lifetime law of the training engines, and the predictions of the methods of Section 6 (ten-seed ensembles for the learned encoders); the age law is the control. It assumes that the test truncation is independent of the lifetime, which the control checks.

Table F.21: Cross-validated age-conditional certificates (nats, 95% bootstrap interval); $\bar{h}(T \mid t)$: mean conditional entropy. Sensors: raw sensors (engines) or charging features (cells); Capacity: reference predictor on the discharge capacity (cells).

Subset	Units	$\bar{h}(T \mid t)$	Age only (control)	Ridge HI	PCA-HI	Level only	Slope only	Sensors	Capacity (ref.)
FD001	100	5.11	-0.11 [-0.28, 0.08]	0.31 [0.20, 0.44]	0.25 [0.13, 0.37]	0.07 [-0.10, 0.24]	0.17 [0.01, 0.32]	0.22 [0.06, 0.38]	–
FD002	260	5.11	-0.11 [-0.22, -0.01]	0.21 [0.13, 0.30]	0.19 [0.11, 0.27]	0.11 [0.03, 0.21]	0.11 [0.01, 0.20]	0.16 [0.08, 0.25]	–
FD003	100	5.64	-0.22 [-0.37, -0.03]	0.10 [-0.05, 0.28]	0.08 [-0.07, 0.25]	0.06 [-0.09, 0.24]	0.00 [-0.13, 0.16]	0.20 [0.04, 0.40]	–
FD004	249	5.52	-0.12 [-0.24, -0.00]	0.18 [0.06, 0.31]	0.02 [-0.11, 0.14]	0.14 [0.01, 0.28]	0.02 [-0.10, 0.14]	0.25 [0.13, 0.38]	–
HUST	77	7.11	-0.02 [-0.14, 0.11]	0.30 [0.19, 0.43]	0.52 [0.42, 0.64]	0.18 [0.06, 0.31]	0.11 [0.01, 0.22]	0.67 [0.57, 0.80]	0.59 [0.48, 0.70]
MATR	169	6.94	-0.27 [-0.46, -0.05]	0.32 [0.17, 0.51]	0.34 [0.17, 0.54]	0.24 [0.07, 0.44]	-0.03 [-0.21, 0.19]	0.61 [0.42, 0.79]	0.29 [0.09, 0.49]
Tongji	88	5.99	0.05 [-0.04, 0.15]	0.60 [0.47, 0.76]	0.52 [0.37, 0.67]	0.33 [0.22, 0.45]	0.27 [0.17, 0.38]	0.91 [0.76, 1.05]	0.40 [0.28, 0.52]

Table F.22: Squeeze of the ridge HI: monotonicity and certified information (nats).

Subset		$\gamma = 0.0$	$\gamma = 0.5$	$\gamma = 0.9$	$\gamma = 0.97$	$\gamma = 0.99$	$\gamma = 0.995$	Mon_∞ (predicted)
FD001	Mon	0.24	0.34	0.54	0.56	0.55	0.55	0.55
	certificate	0.31	0.28	0.21	0.09	-0.06	-0.08	
FD002	Mon	0.19	0.28	0.54	0.58	0.58	0.58	0.61
	certificate	0.21	0.20	0.18	0.11	0.00	-0.06	
FD003	Mon	0.16	0.22	0.40	0.41	0.42	0.42	0.41
	certificate	0.10	0.10	0.10	0.07	0.02	-0.07	
FD004	Mon	0.13	0.18	0.36	0.41	0.41	0.41	0.41
	certificate	0.17	0.18	0.16	0.13	0.07	0.00	
HUST	Mon	0.32	0.53	0.80	0.81	0.81	0.81	0.82
	certificate	0.30	0.28	0.22	0.12	0.04	-0.00	
MATR	Mon	0.39	0.46	0.50	0.48	0.48	0.48	0.49
	certificate	0.32	0.28	0.17	-0.02	-0.14	-0.20	
Tongji	Mon	0.65	0.70	0.73	0.73	0.73	0.73	0.70
	certificate	0.62	0.54	0.33	0.16	0.09	-0.00	

Table F.23: Direction of a scalar linear HI: range of stage-optimal angles, best static angle, and worst-stage RMSE ratios to the free direction.

Subset	stages	optimal angle [deg]	static angle [deg]	static / free	level only / free	velocity only / free
FD001	life fraction	43–70	70	1.019	1.18	1.07
FD001	age	33–69	60	1.021	1.20	1.18
FD002	life fraction	18–66	55	1.002	1.13	1.05
FD002	age	21–56	45	1.037	1.19	1.29
FD003	life fraction	11–33	20	1.003	1.05	1.16
FD003	age	6–35	25	1.025	1.08	1.38
FD004	life fraction	1–20	5	1.007	1.02	1.10
FD004	age	9–37	15	1.018	1.05	1.33
HUST	life fraction	3–80	70	1.090	1.13	1.10
HUST	age	0–57	20	1.184	1.08	1.54
MATR	life fraction	42–88	65	1.000	1.06	1.01
MATR	age	0–10	0	1.012	1.01	1.42
Tongji	life fraction	57–89	90	1.031	1.35	1.03
Tongji	age	13–55	40	1.068	1.12	1.18

Table F.24: RMSE of a linear read-out of the life fraction. Blind: without normalisation by regime (engines) or by the cell's own early cycles (cells).

Subset	Regimes	Blind	Normalised	Regime-specific weights
FD001	1	0.139	0.139	0.139
FD002	6	0.148	0.137	0.136
FD003	1	0.156	0.156	0.156
FD004	6	0.199	0.165	0.164
HUST	one per cell	0.136	0.115	–
MATR	one per cell	0.179	0.142	–
Tongji	one per cell	0.190	0.167	–

Table F.25: Age-conditional certificates (nats) of the benchmark methods on the official test engines. †: the age-only control fails.

Method	FD001	FD002	FD003†	FD004
$h(T \mid t)$	5.09	5.07	5.61	5.47
Age law (control)	0.07 [-0.04, 0.20]	0.01 [-0.07, 0.09]	0.33 [0.23, 0.44]	0.05 [-0.02, 0.13]
PCA-HI	0.59 [0.49, 0.72]	0.29 [0.21, 0.39]	0.67 [0.54, 0.85]	0.26 [0.19, 0.35]
Geo-opt HI	0.58 [0.48, 0.71]	0.23 [0.13, 0.36]	1.10 [0.94, 1.30]	0.66 [0.55, 0.76]
Ridge HI	0.41 [0.26, 0.58]	0.27 [0.17, 0.39]	0.90 [0.77, 1.05]	0.62 [0.54, 0.71]
β-VAE	0.59 [0.47, 0.72]	0.24 [0.16, 0.34]	0.49 [0.35, 0.65]	0.22 [0.14, 0.31]
VIB	0.15 [0.01, 0.32]	0.06 [-0.01, 0.14]	0.57 [0.45, 0.74]	0.11 [0.02, 0.22]
PGIB, integrating (10 seeds)	0.14 [-0.02, 0.32]	0.07 [-0.01, 0.16]	0.49 [0.37, 0.63]	0.19 [0.08, 0.30]
PGIB-KF (10 seeds)	0.61 [0.45, 0.82]	0.28 [0.18, 0.37]	1.04 [0.87, 1.22]	0.48 [0.39, 0.56]
PGIB-KFR (10 seeds)	0.43 [0.28, 0.62]	0.28 [0.20, 0.37]	0.84 [0.73, 0.98]	0.59 [0.51, 0.69]

References

[1] X.-S. Si, W. Wang, C.-H. Hu, D.-H. Zhou, Remaining useful life estimation – a review on the statistical data driven approaches, Eur. J. Oper. Res. 213 (2011) 1–14. https://doi.org/10.1016/j.ejor.2010.11.018

[2] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, J. Lin, Machinery health prognostics: a systematic review from data acquisition to RUL prediction, Mech. Syst. Signal Process. 104 (2018) 799–834. https://doi.org/10.1016/j.ymssp.2017.11.016

[3] K. Javed, R. Gouriveau, N. Zerhouni, State of the art and taxonomy of prognostics approaches, trends of prognostics applications and open issues towards maturity at different technology readiness levels, Mech. Syst. Signal Process. 94 (2017) 214–236. https://doi.org/10.1016/j.ymssp.2017.01.050

[4] E. Zio, Prognostics and health management (PHM): where are we and where do we (need to) go in theory and practice, Reliab. Eng. Syst. Saf. 218 (2022) 108119. https://doi.org/10.1016/j.ress.2021.108119

[5] L. Liao, F. Köttig, Review of hybrid prognostics approaches for remaining useful life prediction of engineered systems, and an application to battery life prediction, IEEE Trans. Reliab. 63 (2014) 191–207. https://doi.org/10.1109/TR.2014.2299152

[6] X. Li, Q. Ding, J.-Q. Sun, Remaining useful life estimation in prognostics using deep convolution neural networks, Reliab. Eng. Syst. Saf. 172 (2018) 1–11. https://doi.org/10.1016/j.ress.2017.11.021

[7] O. Fink, Q. Wang, M. Svensén, P. Dersin, W.-J. Lee, M. Ducoffe, Potential, challenges and future directions for deep learning in prognostics and health management applications, Eng. Appl. Artif. Intell. 92 (2020) 103678. https://doi.org/10.1016/j.engappai.2020.103678

[8] G.A. Whitmore, Estimating degradation by a Wiener diffusion process subject to measurement error, Lifetime Data Anal. 1 (1995) 307–319. https://doi.org/10.1007/BF00985762

[9] M.-L.T. Lee, G.A. Whitmore, Threshold regression for survival analysis: modeling event times by a stochastic process reaching a boundary, Stat. Sci. 21 (2006) 501–513. https://doi.org/10.1214/088342306000000330

[10] L. Wang, S. Chen, Z. Cai, W. Liu, Z. Ye, Cross-cycle health indicator construction and state of health prediction of lithium-ion batteries by cross recurrence quantification analysis, J. Reliab. Sci. Eng. 2 (2026) 035302. https://doi.org/10.1088/3050-2454/ae8dd1

[11] H. Gu, Y. Li, L. Jiang, Z. Luo, A generalized approach for remaining useful life prediction considering the multi-source uncertainty of mechanical systems, J. Reliab. Sci. Eng. 1 (2025) 025301. https://doi.org/10.1088/3050-2454/add5fa

[12] S. Bisoyi, A. Kumar Rathi, S. Mahato, Reliability assessment of remaining useful life predictions of roller bearings using deep learning models, J. Reliab. Sci. Eng. 2 (2026) 025302. https://doi.org/10.1088/3050-2454/ae61b4

[13] T. Aven, The reliability science: its foundation and link to risk science and other sciences, Reliab. Eng. Syst. Saf. 215 (2021) 107863. https://doi.org/10.1016/j.ress.2021.107863

[14] C.-P. Sun, Reliability science and engineering: a catalyst for engineering innovation, J. Reliab. Sci. Eng. 1 (2025) 020201. https://doi.org/10.1088/3050-2454/adbf22

[15] J. Coble, J.W. Hines, Identifying optimal prognostic parameters from data: a genetic algorithms approach, in: Annu. Conf. of the Prognostics and Health Management Society, vol. 1, 2009.

[16] K. Javed, R. Gouriveau, N. Zerhouni, P. Nectoux, A feature extraction procedure based on trigonometric functions and cumulative descriptors to enhance prognostics modeling, in: Proc. IEEE Conf. on Prognostics and Health Management, 2013, pp. 1–7. https://doi.org/10.1109/ICPHM.2013.6621413

[17] B. Zhang, L. Zhang, J. Xu, Degradation feature selection for remaining useful life prediction of rolling element bearings, Qual. Reliab. Eng. Int. 32 (2016) 547–554. https://doi.org/10.1002/qre.1771

[18] F. Qian, G. Niu, Remaining useful life prediction using ranking mutual information based monotonic health indicator, in: Proc. Prognostics and System Health Management Conf., 2015, pp. 1–5. https://doi.org/10.1109/PHM.2015.7380042

[19] L. Guo, N. Li, F. Jia, Y. Lei, J. Lin, A recurrent neural network based health indicator for remaining useful life prediction of bearings, Neurocomputing 240 (2017) 98–109. https://doi.org/10.1016/j.neucom.2017.02.045

[20] D.P. Kingma, M. Welling, Auto-encoding variational Bayes, in: Proc. Int. Conf. on Learning Representations, 2014. arXiv:1312.6114

[21] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, A. Lerchner, β-VAE: learning basic visual concepts with a constrained variational framework, in: Proc. Int. Conf. on Learning Representations, 2017.

[22] K.A. Severson, et al., Data-driven prediction of battery cycle life before capacity degradation, Nat. Energy 4 (2019) 383–391. https://doi.org/10.1038/s41560-019-0356-8

[23] A. Saxena, J. Celaya, E. Balaban, K. Goebel, B. Saha, S. Saha, M. Schwabacher, Metrics for evaluating performance of prognostic techniques, in: Proc. Int. Conf. on Prognostics and Health Management, 2008, pp. 1–17. https://doi.org/10.1109/PHM.2008.4711436

[24] J.L. Folks, R.S. Chhikara, The inverse Gaussian distribution and its statistical application – a review, J. R. Stat. Soc. B 40 (1978) 263–275. https://doi.org/10.1111/j.2517-6161.1978.tb01039.x

[25] R.S. Chhikara, J.L. Folks, *The Inverse Gaussian Distribution: Theory, Methodology, and Applications*, Marcel Dekker, New York, 1989.

[26] M.C.K. Tweedie, Statistical properties of inverse Gaussian distributions. I, Ann. Math. Stat. 28 (1957) 362–377. https://doi.org/10.1214/aoms/1177706964

[27] S. Tang, C. Yu, X. Wang, X. Guo, X. Si, Remaining useful life prediction of lithium-ion batteries based on the Wiener process with measurement error, Energies 7 (2014) 520–547. https://doi.org/10.3390/en7020520

[28] N.Z. Gebraeel, M.A. Lawley, R. Li, J.K. Ryan, Residual-life distributions from component degradation signals: a Bayesian approach, IIE Trans. 37 (2005) 543–557. https://doi.org/10.1080/07408170590929018

[29] X. Wang, Wiener processes with random effects for degradation data, J. Multivariate Anal. 101 (2010) 340–351. https://doi.org/10.1016/j.jmva.2008.12.007

[30] X.-S. Si, W. Wang, C.-H. Hu, D.-H. Zhou, M.G. Pecht, Remaining useful life estimation based on a nonlinear diffusion degradation process, IEEE Trans. Reliab. 61 (2012) 50–67. https://doi.org/10.1109/TR.2011.2182221

[31] K. Medjaher, D.A. Tobon-Mejia, N. Zerhouni, Remaining useful life estimation of critical components with application to bearings, IEEE Trans. Reliab. 61 (2012) 292–302. https://doi.org/10.1109/TR.2012.2194175

[32] Z.-S. Ye, M. Xie, Stochastic modelling and analysis of degradation for highly reliable products, Appl. Stoch. Models Bus. Ind. 31 (2015) 16–32. https://doi.org/10.1002/asmb.2063

[33] Z. Zhang, X. Si, C. Hu, Y. Lei, Degradation data analysis and remaining useful life estimation: a review on Wiener-process-based methods, Eur. J. Oper. Res. 271 (2018) 775–796. https://doi.org/10.1016/j.ejor.2018.02.033

[34] S. Sankararaman, K. Goebel, Why is the remaining useful life prediction uncertain?, in: Annu. Conf. of the Prognostics and Health Management Society, vol. 5, 2013. https://doi.org/10.36001/phmconf.2013.v5i1.2263

[35] S. Sankararaman, Significance, interpretation, and quantification of uncertainty in prognostics and remaining useful life prediction, Mech. Syst. Signal Process. 52–53 (2015) 228–247. https://doi.org/10.1016/j.ymssp.2014.05.029

[36] E. Hüllermeier, W. Waegeman, Aleatoric and epistemic uncertainty in machine learning: an introduction to concepts and methods, Mach. Learn. 110 (2021) 457–506. https://doi.org/10.1007/s10994-021-05946-3

[37] A. Kendall, Y. Gal, What uncertainties do we need in Bayesian deep learning for computer vision?, in: Advances in Neural Information Processing Systems, vol. 30, 2017. arXiv:1703.04977

[38] B. Lakshminarayanan, A. Pritzel, C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in: Advances in Neural Information Processing Systems, vol. 30, 2017. arXiv:1612.01474

[39] C.E. Shannon, A mathematical theory of communication, Bell Syst. Tech. J. 27 (1948) 379–423, 623–56. https://doi.org/10.1002/j.1538-7305.1948.tb01338.x

[40] T.M. Cover, J.A. Thomas, *Elements of Information Theory, 2nd ed.*, Wiley-Interscience, Hoboken, NJ, 2006.

[41] N. Tishby, F.C. Pereira, W. Bialek, The information bottleneck method, arXiv:physics/0004057 (2000).

[42] A.A. Alemi, I. Fischer, J.V. Dillon, K. Murphy, Deep variational information bottleneck, in: Proc. Int. Conf. on Learning Representations, 2017. arXiv:1612.00410

[43] B. Poole, S. Ozair, A. van den Oord, A.A. Alemi, G. Tucker, On variational bounds of mutual information, in: Proc. Int. Conf. on Machine Learning (PMLR vol 97), 2019, pp. 5171–5180.

[44] A. Achille, S. Soatto, Emergence of invariance and disentanglement in deep representations, J. Mach. Learn. Res. 19 (2018) 1–34.

[45] A. Kolchinsky, B.D. Tracey, D.H. Wolpert, Nonlinear information bottleneck, Entropy 21 (2019) 1181. https://doi.org/10.3390/e21121181

[46] A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, A. Kolchinsky, B.D. Tracey, D.D. Cox, On the information bottleneck theory of deep learning, J. Stat. Mech.: Theory Exp. 2019 (2019) 124020. https://doi.org/10.1088/1742-5468/ab3985

[47] O. Vasicek, A test for normality based on sample entropy, J. R. Stat. Soc. B 38 (1976) 54–59. https://doi.org/10.1111/j.2517-6161.1976.tb01566.x

[48] A. Kraskov, H. Stögbauer, P. Grassberger, Estimating mutual information, Phys. Rev. E 69 (2004) 066138. https://doi.org/10.1103/PhysRevE.69.066138

[49] L. Paninski, Estimation of entropy and mutual information, Neural Comput. 15 (2003) 1191–1253. https://doi.org/10.1162/089976603321780272

[50] D. McAllester, K. Stratos, Formal limitations on the measurement of mutual information, in: Proc. Int. Conf. on Artificial Intelligence and Statistics (PMLR vol 108), 2020, pp. 875–884.

[51] H.L. Van Trees, *Detection, Estimation, and Modulation Theory, Part I*, Wiley, New York, 1968.

[52] R.D. Gill, B.Y. Levit, Applications of the van Trees inequality: a Bayesian Cramér–Rao bound, Bernoulli 1 (1995) 59–79. https://doi.org/10.2307/3318681

[53] P. Tichavský, C.H. Muravchik, A. Nehorai, Posterior Cramér–Rao bounds for discrete-time nonlinear filtering, IEEE Trans. Signal Process. 46 (1998) 1386–1396. https://doi.org/10.1109/78.668800

[54] Anonymous, *Reference withheld for double-blind review*, , 2026.

[55] Anonymous, *Reference withheld for double-blind review*, , 2026.

[56] Anonymous, *Reference withheld for double-blind review*, , 2026.

[57] C.M. Bishop, *Pattern Recognition and Machine Learning*, Springer, New York, 2006.

[58] R.E. Kalman, A new approach to linear filtering and prediction problems, J. Basic Eng. 82 (1960) 35–45. https://doi.org/10.1115/1.3662552

[59] J. Durbin, S.J. Koopman, *Time Series Analysis by State Space Methods, 2nd ed.*, Oxford University Press, Oxford, 2012.

[60] A.C. Harvey, *Forecasting, Structural Time Series Models and the Kalman Filter*, Cambridge University Press, Cambridge, 1989.

[61] G. Ma, et al., Real-time personalized health status prediction of lithium-ion batteries using deep transfer learning, Energy Environ. Sci. 15 (2022) 4083–4094. https://doi.org/10.1039/D2EE01676A

[62] P.M. Attia, et al., Closed-loop optimization of fast-charging protocols for batteries with machine learning, Nature 578 (2020) 397–402. https://doi.org/10.1038/s41586-020-1994-5

[63] J. Zhu, et al., Data-driven capacity estimation of commercial lithium-ion batteries from voltage relaxation, Nat. Commun. 13 (2022) 2261. https://doi.org/10.1038/s41467-022-29837-w

[64] R. Tan, et al., BatteryLife: a comprehensive dataset and benchmark for battery life prediction, in: Proc. 31st ACM SIGKDD Conf. on Knowledge Discovery and Data Mining, vol. 2, 2025, pp. 5789–5800. https://doi.org/10.1145/3711896.3737372

[65] X. Bouthillier, et al., Accounting for variance in machine learning benchmarks, Proc. Mach. Learn. Syst. 3 (2021) 747–769.

[66] B. Efron, R.J. Tibshirani, *An Introduction to the Bootstrap*, Chapman and Hall, New York, 1993.

[67] F. Mauthe, C. Braun, J. Raible, P. Zeiler, M.F. Huber, Overview of publicly available degradation data sets for tasks within prognostics and health management, arXiv preprint (2024). arXiv:2403.13694

[68] A.J. Stam, Some inequalities satisfied by the quantities of information of Fisher and Shannon, Inf. Control 2 (1959) 101–112. https://doi.org/10.1016/S0019-9958(59)90348-1

[69] R. Durrett, *Probability: Theory and Examples, 4th ed.*, Cambridge University Press, Cambridge, 2010.

[70] C. Genest, J.V. Zidek, Combining probability distributions: a critique and an annotated bibliography, Stat. Sci. 1 (1986) 114–135. https://doi.org/10.1214/ss/1177013825

[71] G.E. Hinton, Training products of experts by minimizing contrastive divergence, Neural Comput. 14 (2002) 1771–1800. https://doi.org/10.1162/089976602760128018

[72] W.Q. Meeker, L.A. Escobar, *Statistical Methods for Reliability Data*, Wiley, New York, 1998.

[73] T. Gneiting, A.E. Raftery, Strictly proper scoring rules, prediction, and estimation, J. Am. Stat. Assoc. 102 (2007) 359–378. https://doi.org/10.1198/016214506000001437

[74] B. Saha, K. Goebel, Battery data set, NASA Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, 2007.

[75] W. He, N. Williard, M. Osterman, M. Pecht, Prognostics of lithium-ion batteries based on Dempster–Shafer theory and the Bayesian Monte Carlo method, J. Power Sources 196 (2011) 10314–10321. https://doi.org/10.1016/j.jpowsour.2011.08.040

[76] P. Nectoux, R. Gouriveau, K. Medjaher, E. Ramasso, B. Chebel-Morello, N. Zerhouni, C. Varnier, PRONOSTIA: an experimental platform for bearings accelerated degradation tests, in: Proc. IEEE Int. Conf. on Prognostics and Health Management (Denver, CO), 2012, pp. 1–8.

[77] J. Lee, H. Qiu, G. Yu, J. Lin, Rexnord Technical Services, Bearing data set, IMS, University of Cincinnati, NASA Prognostics Data Repository, NASA Ames Research Center, Moffett Field, CA, 2007.

[78] H. Qiu, J. Lee, J. Lin, G. Yu, Wavelet filter-based weak signature detection method and its application on rolling element bearing prognostics, J. Sound Vib. 289 (2006) 1066–1090. https://doi.org/10.1016/j.jsv.2005.03.007

[79] A. Saxena, K. Goebel, D. Simon, N. Eklund, Damage propagation modeling for aircraft engine run-to-failure simulation, in: Proc. Int. Conf. on Prognostics and Health Management (Denver, CO), 2008, pp. 1–9. https://doi.org/10.1109/PHM.2008.4711414